Planetary Motion and the Three-Body Problem

The Mathematics Behind the "Simulator" Computer Program

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This document describes planetary motion and aspects of the three-body problem, insofar as these are accessible at a basic level. The mathematical investigations are supported by experiments using the computer program “Simulator,” which enables the simulation of n-body systems in the universe.

The document is intended to serve as a resource for advanced mathematics instruction at the middle school level, whether for activities outside the mandatory curriculum or for individual projects by interested students. It is particularly suitable as an introduction to the theory of ordinary differential equations.

The entire series of topics related to “Simulator” includes:

- *The chaotic properties of logistic growth*
- *The oval billiard table and periodic orbits*
- *Newton’s iteration and the complex unit roots*
- *Iteration of quadratic functions in the complex plane*
- *Numerical methods and coupled pendulums*
- ***Planetary motion and the three-body problem***
- *Strange attractors and Edward Lorenz’s weather forecast*
- *Fractal sets and Lindenmayer systems*
- *The history of chaos theory*
- *Programming your own dynamic systems in the “Simulator”*

Each topic is covered in a separate document.

The “Simulator” computer program allows you to simulate simple dynamic systems and experiment with them. The code is publicly available on GitHub, as is a Microsoft Installer version. The link is: https://github.com/HermannBiner/Simulator. The following documentation is included in the “Simulator” in both German and English:

- *Mathematical documentation with examples and exercises*
- *Technical documentation with a detailed description of the functionality*
- *User manual with examples*
- *Version history*

Contents

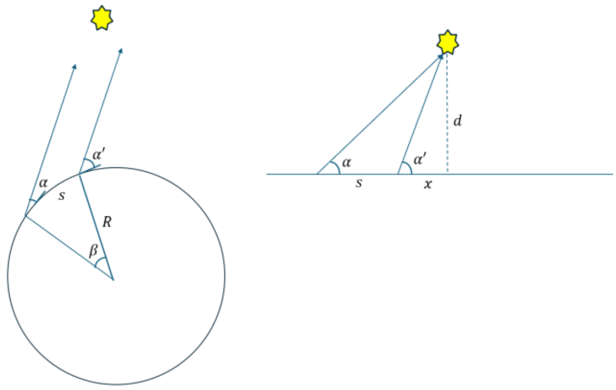
1. Space and Time

We have a fairly good idea of the space that surrounds us, at least within our immediate surroundings. Even though we live on a globe, it feels (locally) Euclidean. That is why there are still people today who claim that the Earth is flat, without specifying what its edge should look like. "I've been to various places on Earth, and apart from local unevenness, it was always flat." A fly crawling around on a very large saddle surface would say the same thing.

This makes it all the more remarkable that the ancient Greeks already viewed the Earth as a sphere. Several observations were more consistent with a spherical Earth: When ships disappear over the horizon, this happens gradually. First the hull disappears, then the mast. Conversely, tall buildings appear on the horizon first when a ship approaches the mainland. During lunar eclipses, the Earth always casts a round shadow, and that is only possible with a spherical shape. Theoretically, one could check the sum of the angles in a large triangle on Earth (and the fly on the saddle surface could do the same). If the Earth is a sphere, this sum is greater than 180° (for the fly, it would be less than 180°). However, for this deviation to be at least 1°, an equilateral triangle with a side length of about 1,276 km would be required on Earth. This was not feasible for the Greeks.

One could use the fact that, in the northern hemisphere, the North Star always forms the same angle with the horizon from a given position on Earth throughout the year and regardless of the time of day (or night), because the Earth's axis, apart from a small precession , is always pointed toward it. This angle depends on the latitude of the position. There are two possible explanations for this:

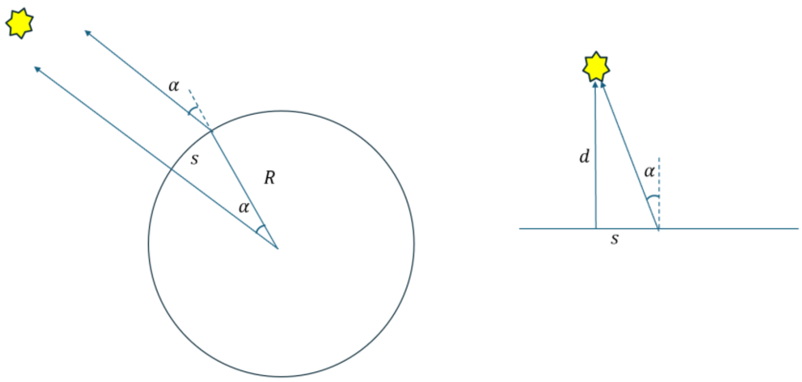
Either the Earth is a sphere and the rays of sight to the North Star are parallel, or the North Star is much closer to Earth than expected.



Position of the North Star: Left in the case of a spherical Earth, right in the case of a flat Earth

The Greek mathematician Eratosthenes (c. 275–193 BCE) conducted an experiment on this. He did not observe the North Star, but rather the angle of the sun's shadow in Alexandria and Syene (modern-day Aswan). This had to be measured at both locations on the same day and at the same time. When the sun was at its highest point on the summer solstice—that is, the longest day on June 21—no shadow was visible in a deep well in Syene. This means the sun was directly overhead in Syene. On the same day and at the same time, a stick in Alexandria cast a shadow that formed an angle of $\alpha = 7.2°$ with the sun. When the sun is at its highest point in both locations, this difference arises solely from the different latitudes of these positions and is identical to the corresponding central angle at the Earth's center. In fact, the latitude of Syene is 24°05' N and that of Alexandria is 31°13' N. That is a difference of 7.15°.

We do not know exactly how Eratosthenes measured the corresponding distance between Alexandria and Syene, although he only had to account for the distance between the latitudes, meaning he should have measured along a meridian. Apparently, he estimated this distance and arrived at 5,000 "stadiums." Eratosthenes likely used the Egyptian stadium, which measures 157.5 m. Thus, $s = 5000 \cdot 157.5 \text{ m} = 787.5 \text{ km}$.

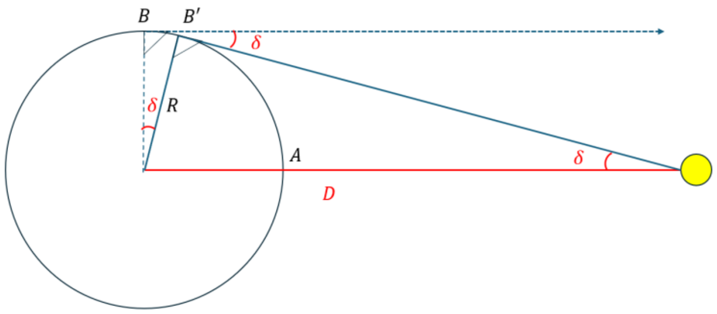


Eratosthenes' experiment: on the left for a sphere, on the right for a flat Earth

Now, for the circumference of the Earth, we have $U = s \cdot \frac{360}{\alpha}$. This gives us $U = 39'375 \text{ km}$ and for the radius of the Earth $R = \frac{U}{2\pi} \approx 6'267 \text{ km}$. This is a surprisingly good estimate for the value that we now know to be correct: $6'371 \text{ km}$.

Suppose the Earth were flat. Then the angle of 7.2° would result because the Sun is close to the Earth. Given the measured values for the Earth–Sun distance, this would yield the improbable value $d = \frac{s}{\tan\alpha} = 6'233.7\text{ km}$! By way of comparison: The Earth–Moon distance is 384,400 km.

The Greeks also had an estimate for this distance. One could apply a method like that used for estimating the Earth's radius. Let's conduct a thought experiment:



Thought experiment for measuring parallax

For an observer at the point A, the Moon is perpendicular to their plane of observation. A second observer looks for a point at which the Moon is exactly on the horizon at the same time. If the Moon were very far away, this would be the point B. The dashed line would then be parallel to the line between the center of the Earth and the Moon in the sketch above. However, if the Moon is closer to Earth, it is already on the horizon for an observer at point B'. This results in what is known as *parallax*. This is the deviation δ from the parallel direction. However, with this method for measuring δ, there is the problem that the measurement must be taken simultaneously at A and B'. If one were to allow oneself to be "transported" by the Earth's rotation from A to B', one must account for the Moon's displacement during that time.

The Greek astronomer Hipparchus of Nicaea (c. 190–120 BCE) calculated the parallax for the Moon. However, his writings have been lost, and it is not known how he performed the calculation. Perhaps he used observations of a solar eclipse that occurred shortly before his time, specifically on March 14, 190 BC: In the Hellespont (today the Dardanelles), the Sun was completely obscured by the Moon. In Alexandria, only about 4/5 of it was covered. Hipparchus thus calculated an estimate of: $\delta \approx 58' = 0.9667°$. It then follows that: $D = {}^{R}/_{\tan\delta} = R \cdot 59.3$, where R is the Earth's radius. This comes remarkably close to the actual value $D = R \cdot 60.3$.

We have gone to some effort so far to show that approximations for the Earth's radius and the Earth–Moon distance can be calculated using elementary methods. The reason for this is that Newton needed these measurements when discovering the law of gravity.

A problem arose during the calculations: the experiments had to be conducted at different locations *at the same time*. Eratosthenes solved this problem with the help of the summer solstice. Without clocks, however, we have no natural "sense" of time. We have the greatest difficulty estimating how long an hour lasts. At a party, it seems to fly by in no time. At the dentist, it lasts forever. It is a mystery to us. We can distinguish between past, present, and future. But the present "flows steadily forward" without us being able to influence it.

Suppose we have no clocks and are living in the time of the ancient Greeks. If we are outside during the day and the weather is good, we could use the position of the sun to tell the time. The phases of the moon would help us divide the year into months. If we were to drive a vertical stick into the ground, we could mark the tip of the shadow throughout the year and thus estimate both the time of day and the season in subsequent years. If we need fixed points in time that are independent of geographical location, things get difficult. One such reference point could be the summer solstice. Above the tropics, this day is the same for every location and can be determined locally as the day when the sun reaches its highest point in the sky. It also rises at the easternmost point on the horizon, and this point can be marked in the landscape. Stonehenge is one such example.

The axis of Stonehenge is aligned with the summer solstice point

The starry sky and its “movement” relative to Earth can also serve as a calendar for us. Where does this movement come from? If you look at the starry sky, you can “see” that it revolves around the North Star.

Does the starry sky rotate around an axis through the North Star?

The Sun, Moon, and planets also move across the sky. The Earth, on the other hand, is at rest: no one on Earth has ever felt a “headwind” or acceleration forces. Yes, one could still argue this today in certain circles.

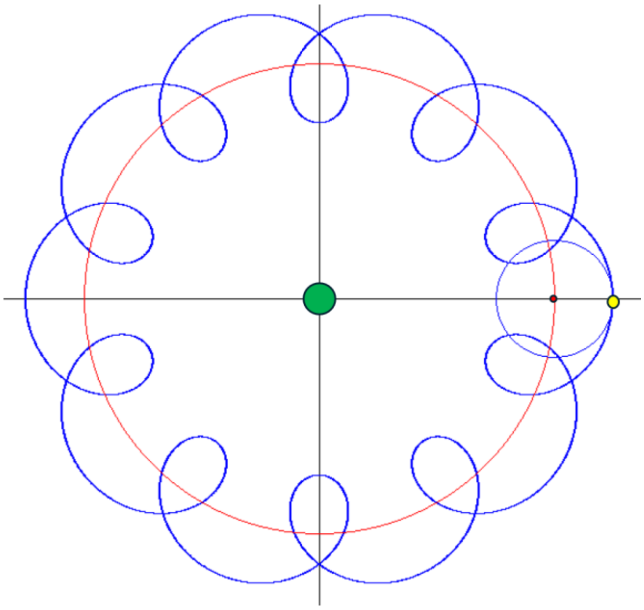
What is certain, however, is that the relative distances between the fixed stars remain constant, at least in the short term. In contrast, the planets move relative to the fixed starry sky, and at different speeds. As seen from Earth, they sometimes even appear to move backward, namely when, in reality, Earth is overtaking a slower planet (e.g., Mars).

Time as a phenomenon in itself was not questioned. It was clear that clocks run at the same speed at all points in the universe, which was conceived as a three-dimensional Euclidean space. Since Albert

Einstein examined the concept of “simultaneity” and postulated the speed of light as a global constant independent of the observer’s frame of reference, everything became much more complex: space and time form a unity, four-dimensional spacetime. This is no longer Euclidean, but is “curved” by the presence of matter. Clocks run slower in a gravitational field. This is a revolution that has upended our worldview at least as much as the realization that the Earth might indeed move and that it does not lie at the centre of the solar system.

2. The Centre of the Planetary System

Aristotle (384–322 BCE) postulated a geocentric worldview. Claudius Ptolemy (c. 100–160) refined this model using epicycles: the planets moved in small circles whose centres moved along a main circle around the Earth. This also explained the sometimes-retrograde motion of the planets. He refined this model to the point where it agreed very well with observations.



The yellow planet revolves around the small circle, whose centre simultaneously rotates on the large, red circle. At the centre is the Earth

Nicolaus Copernicus (1473–1543) proposed the heliocentric model as an alternative, in which the Sun is at the center, the Earth is an ordinary planet that orbits the Sun and rotates simultaneously. This allowed him to elegantly explain the sometimes-retrograde motion of the planets. However, his model was no more precise than the geocentric model with its epicycles.

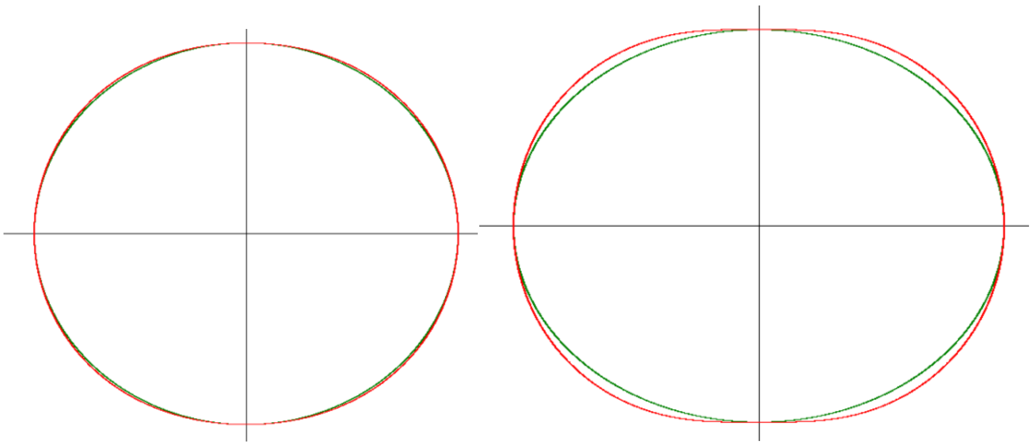
So now there were two models, and precise observations were needed to determine which was better suited to explaining the positions of the planets. Tycho Brahe (1546–1601) provided this data. He observed the starry sky for decades with a precision—unprecedented at the time—of up to one arcminute. It is important to remember that at that time, the positions of the planets could only be determined using angular measurements. The true distance between Earth and the Sun was unknown; only the Sun’s position relative to Earth was known, expressed in terms of angles. For these angular measurements, Tycho Brahe used spherical coordinates. He defined the ecliptic as the reference plane and the vernal equinox as the origin on the ecliptic. This is the intersection of the ecliptic and the equatorial plane, which the Sun crosses at the beginning of spring on March 21. This defined the reference system for the positions, with the actual distances determined relative to the astronomical unit (the distance between Earth and the Sun). His model was a compromise between geocentric and heliocentric worldviews: the stationary Earth stood at the centre of the universe, orbited by the Moon. The Sun moved once a day around the Earth, and the other planets around the Sun.

Johannes Kepler (1571–1630) collaborated with Tycho Brahe and, after Brahe's death, "inherited" his highly precise data on planetary positions. He attempted to explain this data and employed Copernicus's heliocentric model, including corrections via epicycles. However, even with epicycles, he could not explain a persistent deviation of 8′ for Mars. Finally, he made the groundbreaking discovery: The model matched the observations precisely if one postulated ellipsis for the planetary orbits instead of working with epicycles.

We want to investigate in a computer simulation how an ellipse can be approximated by an epicyclic orbit. An exercise demonstrates that a good approximation for the ellipse with semi-axes a, b is the curve

$$\vec{r}(t) = \frac{a+b}{2}\begin{pmatrix}\cos t\\ \sin t\end{pmatrix} + \frac{a-b}{2}\begin{pmatrix}\cos{(3t)}\\ \sin{(3t)}\end{pmatrix}$$

, at least as long as$a$ and$b$ do not differ too greatly from one another. In the following figure, the ellipse is shown in green and the epicyclic approximation in red.



Left: Ellipse with $a = 1, b = 0.9$. Right: Ellipse with $a = 1, b = 0.8$

On the left, the deviation is small and barely noticeable. On the right, it is clearly visible.

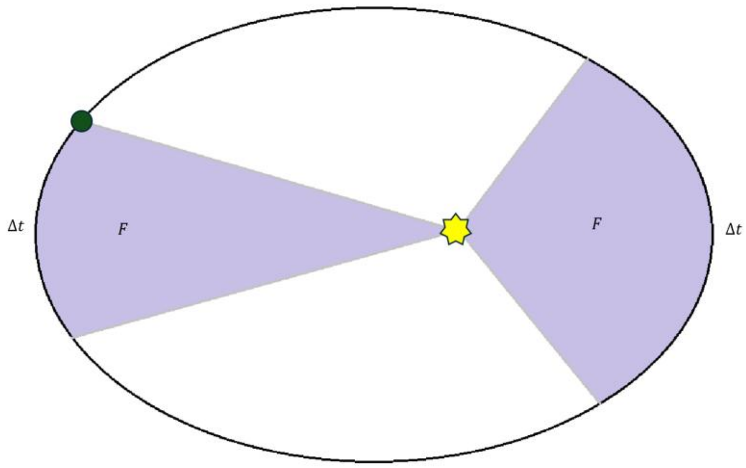
For the planets, the semi-major axes differ very little from one another. The difference is greatest for Mars. If one normalizes the semi-major axis to $a = 1$, then $b = 0.995$ holds. Saturn has the second-largest deviation among the planets known at the time. If one normalizes its semi-major axis to $a = 1$, then $b = 0.9985$ holds. Here, the difference between the ellipse and the epicycle is barely discernible.

Based on his findings, Kepler formulated the following laws:

Theorem 2.1 (*Keplers Laws*)

1) The planets move in ellipses, with the Sun at one of the foci.
2) The line connecting a planet to the Sun sweeps out equal areas in equal intervals of time.
3) If one compares two planets and the semi-major axes (a_1, a_2) of their elliptical orbits, then the following holds: $(T_1/T_2)^2 = (a_1/a_2)^3$, where T_1, T_2 are the times, the planets take to complete one orbit.

□



Keplers First and Second Laws

This model explained the observations precisely, and thus the geocentric worldview was no longer tenable from a scientific perspective. However, it took over 150 years for the scientific method to prevail over religious and philosophical ideas.

An influential advocate of the heliocentric model of the universe was Galileo Galilei (1564–1641). He was a polymath and made contributions in all fields of science and medicine. He developed a telescope with 30x magnification by grinding the lenses himself. With it, he discovered Jupiter's four largest moons. He observed that Venus, like the Moon, had phases, sometimes lying between Earth and the Sun, and sometimes beyond the Sun. This was only possible within the heliocentric model of the universe.

Even though some church officials also accepted the heliocentric model of the universe, this model—and thus its proponents, including Galileo—fell under the ban of the Inquisition in 1616. It was not until 1758 that the Catholic Church lifted the ban on the dissemination of the heliocentric model of the universe.

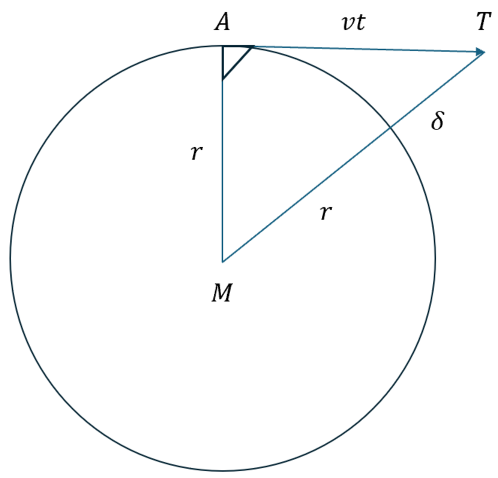
3. The Moon and the Law of Gravity

In the school project *"Numerical Methods and Coupled Pendulums*," we learned about Newton's laws. Stated somewhat freely, they are as follows:

- When a body is in motion, no force is required to maintain its velocity—contrary to earlier beliefs—but rather the body moves "on its own" at a constant velocity
- Force = Reaction Force: If the Earth exerts a force on the Moon, then the Moon exerts the same force on the Earth, but in the opposite direction.
- Force = mass × acceleration

Kepler had already observed that bodies moving in circular orbits have "a tendency to move outward." He hypothesized that the Sun exerts a kind of "magnetic" force on the planets to keep them in their orbits and counteract this "tendency to move outward."

Christian Huygens (1629–1695) analysed this "tendency to move outward" in a clever way. His reasoning was as follows:



Huygens' analysis of centrifugal acceleration

Suppose a body moves with tangential velocity$v$ along a circle centred at M. If it were to continue moving tangentially from the point A, it would have travelled a distance of vt after a time of t. In reality, however, it remains on the circle, so it has simultaneously travelled a distance of δ in the direction of M. In the right-angled triangle MAT, the following holds:

$$\delta = \sqrt{r^2 + (vt)^2} - r$$

Ift is very small, then $vt \ll r$, and thus the approximation (which can be verified by squaring) holds:

$$\sqrt{r^2 + (vt)^2} \approx r + \frac{(vt)^2}{2r}$$

Thus: $\delta \approx \frac{1}{2} \cdot \frac{v^2}{r} \cdot t^2$. Now, Huygens compared this result with free fall, for which, according to Galileo, under uniform acceleration$a$, the following holds for the distance traveled$s$: $s = \frac{1}{2} \cdot a \cdot t^2$.

Thus, he concluded: The centripetal acceleration that keeps the body on a circular path with radius$r$ must be:

$$a = \frac{v^2}{r}$$

Using Newton's law $F = ma$, the required centripetal force that keeps the body on the circular path is thus:

Theorem 3.1

For a body moving in a circular path with constant tangential velocity v to remain on this path, the following centripetal force must act on it:

$$F_p = m\frac{v^2}{r}$$

□

Isaac Newton (1643–1727) formulated not only Newton's laws of motion but also the law of gravitation. He developed calculus and was thus able to derive that Keplers laws follow from the law of gravitation. This was considered one of the greatest achievements in the history of physics.

(Calculus was developed independently of Newton and, in parallel, by Gottlieb Wilhelm Leibniz (1646–1716).)

For the planets to remain in their orbits, a centripetal force had to be at work. Legend has it that a falling apple gave Newton the idea that the same force that caused the apple to fall also acted as a centripetal force on the moon. The concept of "gravitation" was already in use before Newton as "heaviness." Newton then estimated, based on the Moon's orbit, what this gravitational force should look like. It should exactly equal the centripetal force. He postulated that the gravitational force decreases with the square of the distance and tested this hypothesis using the Moon's motion. To do this, he needed only the Earth's radius $R_\oplus$, the acceleration due to gravity g, and the Earth–Moon distance r. Together with the orbital period T of the Moon around the Earth, all these quantities were known:

$$R_\oplus \approx 6.37 \cdot 10^6 \text{ m}$$

$$g \approx 9.81 \ {}^{\text{m}}/_{\text{s}^2}$$

$$r \approx 3.84 \cdot 10^8 \text{ m}$$

$$T \approx 27.32 \text{ days} = 2.36 \cdot 10^6 \text{ s}$$

Let's try to reconstruct this experiment. Let's imagine the Moon as a point mass with mass m_M. If it were on the Earth's surface, the gravitational force acting on it would be

$$F_{R_\oplus} = m_M g$$

If the force decreases proportionally to the square of the distance, the gravitational force on the Moon at a distance r

$$F_r = m_M g \left(\frac{R_\oplus}{r}\right)^2$$

The Moon's velocity is $v = \frac{2\pi r}{T}$. Thus, the required centripetal force is

$$F_p = m_M \frac{4\pi^2 r}{T^2}$$

So, if Newton's assumption is correct, then the following must hold

$$F_r = F_p$$

Here, the mass of the Moon cancels out! Using the above numbers, we get (rounded to two decimal places):

$$\frac{F_r}{m_M} \approx 9.81 \cdot \left(\frac{6.37 \cdot 10^6}{3.84 \cdot 10^8}\right)^2 = 27.00 \cdot 10^{-4}$$

$$\frac{F_p}{m_M} \approx \frac{4\pi^2 \cdot 3.84 \cdot 10^8}{(2.36 \cdot 10^6)^2} = 27.22 \cdot 10^{-4}$$

This agrees very well with the fact that we are working with approximations, including regarding the Moon's orbit. From symmetry, Newton concluded: Since the force is proportional to the Moon's mass, the counterforce must be proportional to the Earth's mass. Thus, the following holds:

Theorem 3.2 (*Law of Gravitation*)

Between two masses m_1, m_2 located at a distance r from each other, the gravitational force acts:

$$F = G\frac{m_1 m_2}{r^2}$$

□

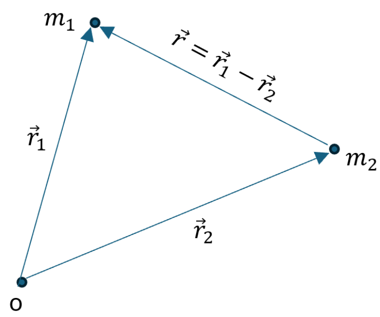
Since Newton knew neither the mass of the Earth nor that of the Moon, he could not determine the gravitational constant $G \approx 6.6743 \cdot 10^{-11} \frac{\mathrm{m}^3}{\mathrm{kg \cdot s^2}}$, but worked solely with proportions. However, the product $GM_\oplus \approx 3.98 \cdot 10^{14}\ \mathrm{m}^3/\mathrm{s}^2$ was known, where $M_\oplus$ is the mass of the Earth. This product is derived from the equation for the gravitational force and a mass m at the Earth's surface: $F = mg = GM_\oplus \frac{m}{R_\oplus^2}$.

The gravitational constant was first approximately determined in 1797 by Henry Cavendish using a gravitational balance.

4. The Two-Body Problem

In the following, we examine the effects of the law of gravitation on two masses moving in each other's gravitational field. We consider two masses m_1, m_2 located at positions $\vec{r}_1(t), \vec{r}_2(t)$ at time t. Here, $\vec{r}_1(0) \neq \vec{r}_2(0)$.

A force acts between the masses $\vec{F} = G\frac{m_1 m_2}{|\vec{r}|^2}$ in the direction of the other mass, where $\vec{r} = \vec{r}_1 - \vec{r}_2 \neq \vec{0}$ is the position difference vector



Two masses in the universe

Based on Newton's second law and the law of gravity, we have the equations of motion:

$$(1)\begin{cases} m_1\ddot{\vec{r}}_1 = -G\dfrac{m_1m_2}{|\vec{r}|^2}\cdot\dfrac{\vec{r}}{|\vec{r}|} \\ m_2\ddot{\vec{r}}_2 = G\dfrac{m_1m_2}{|\vec{r}|^2}\cdot\dfrac{\vec{r}}{|\vec{r}|} \end{cases}$$

Theorem 4.1

For two masses m_1, m_2 at positions $\vec{r}_1, \vec{r}_2$, moving in each other’s gravitational field, the equations of motion are:

$$(2)\begin{cases} \ddot{\vec{r}}_1 = -G\dfrac{m_2}{|\vec{r}|^2}\cdot\dfrac{\vec{r}}{|\vec{r}|} \\ \ddot{\vec{r}}_2 = G\dfrac{m_1}{|\vec{r}|^2}\cdot\dfrac{\vec{r}}{|\vec{r}|} \end{cases}$$

Here,$\vec{r} = \vec{r}_1 - \vec{r}_2$.

□

First, we want to show that the classical conservation laws follow from these equations.

We will first examine the *momentum theorem*. Adding the equations (1) yields:

$$m_1\ddot{\vec{r}}_1 + m_2\ddot{\vec{r}}_2 = \vec{0}$$

Integrating with respect to *t* to find the total momentum of the system yields:

$$m_1\dot{\vec{r}}_1 + m_2\dot{\vec{r}}_2 = \vec{p}$$

Where $\vec{p}$ is constant.

The centre of mass of the system is the mass-weighted average of its position coordinates. If $\vec{r}_s$ is the coordinate of the centre of mass and $M := m_1 + m_2$, then:

$$\vec{r}_s = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{M}$$

From this, the total momentum of the system is:

$$M\dot{\vec{r}}_s = m_1\dot{\vec{r}}_1 + m_2\dot{\vec{r}}_2$$

As a corollary, it follows from $M\dot{\vec{r}}_s = \vec{p} =$ constant that the centre of mass of the system moves at *a constant velocity*.

Now let us examine the *angular momentum theorem*. For the total angular momentum $\vec{L}$ of the system, the following holds:

$$\frac{d}{dt}\vec{L} = \frac{d}{dt}\left(\vec{r}_1 \times m_1\dot{\vec{r}}_1 + \vec{r}_2 \times m_2\dot{\vec{r}}_2\right) = \vec{r}_1 \times m_1\ddot{\vec{r}}_1 + \vec{r}_2 \times m_2\ddot{\vec{r}}_2$$

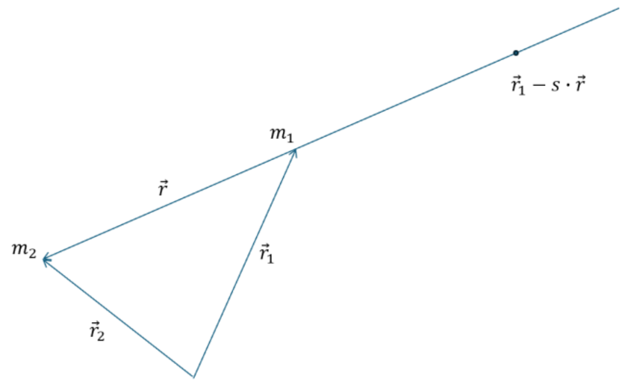
$$= -\vec{r}_1 \times G\frac{m_1m_2}{|\vec{r}|^2}\cdot\frac{\vec{r}}{|\vec{r}|} + \vec{r}_2 \times G\frac{m_1m_2}{|\vec{r}|^2}\cdot\frac{\vec{r}}{|\vec{r}|} = \vec{r} \times G\frac{m_1m_2}{|\vec{r}|^2}\cdot\frac{\vec{r}}{|\vec{r}|} = \vec{0}$$

Thus, the angular momentum $\vec{L}$ of the system is constant.

We will also verify that the *total energy* of the system is constant. To do this, we need the potential energy. This is defined by the work that must be done to move a mass in a force field. Here, the infinitesimal work is the scalar product of the force acting at a point and the infinitesimal distance traveled.

For the force field, we can assume the gravitational field ofm_2 and move the mass m_1 within it. Conversely, we could also assume the gravitational field of $m_1$and move m_2 within it, obtaining the same result for energy conservation. However, one cannot mix the two, otherwise one “counts” the potential energy twice.

Now we normalize the potential energy so that its zero point lies at infinity. If the mass m_1 is at the position $\vec{r}_1$, then its potential energy is equal to the work that must be done to move this mass from its position to infinity. Since we need the scalar product of force and displacement for the work, we choose a displacement opposite to the direction of the force. This is a displacement along the line $\vec{r}_1 - s\cdot\vec{r}, 0 \le s < \infty$.



Calculation of the potential energy of the mass m_1 at the position $\vec{r}_1$ in the gravitational field of m_2

At the point $\vec{r}_1 - s\cdot\vec{r}$, the gravitational force acts

$$\vec{F} = -G\frac{m_1m_2}{|\vec{r}_1 - s\cdot\vec{r} - \vec{r}_2|^2}\cdot\frac{\vec{r}}{|\vec{r}|} = -G\frac{m_1m_2}{|(1-s)\vec{r}|^2}\cdot\frac{\vec{r}}{|\vec{r}|} = -G\frac{m_1m_2\vec{r}}{(1-s)^2|\vec{r}|^3}$$

The infinitesimal path element is: $-\vec{r}ds$. Thus, the potential energy is the integral:

$$E_{pot} = \int_0^\infty G\frac{m_1m_2\vec{r}\cdot\vec{r}}{(1-s)^2|\vec{r}|^3}ds = G\frac{m_1m_2}{|\vec{r}|}\int_0^\infty \frac{ds}{(1-s)^2}$$

As can be verified by calculation, the following holds:

$$\frac{d}{ds}\frac{1}{1-s}=\frac{1}{(1-s)^2}$$

Thus,

$$E_{pot}=-G\frac{m_1 m_2}{|\vec{r}|}$$

Since the denominator contains only the magnitude of the distance between the masses m_1 and m_2, one would obtain the same result if, instead of moving m_1 in the gravitational field of m_2, one were to move m_2 in the gravitational field of m_1.

For the total energy, we then have

$$E=\frac{1}{2}m_1\left|\dot{\vec{r}}_1\right|^2+\frac{1}{2}m_2\left|\dot{\vec{r}}_2\right|^2-G\frac{m_1 m_2}{|\vec{r}|}$$

Now let's consider $\vec{r}_1, \vec{r}_2$ as a function of time t, and calculate:

$$\frac{d}{dt}E=m_1\dot{\vec{r}}_1\cdot\ddot{\vec{r}}_1+m_2\dot{\vec{r}}_2\cdot\ddot{\vec{r}}_2+Gm_1m_2\frac{\vec{r}\cdot\dot{\vec{r}}}{|\vec{r}|^3}$$

We set $\gamma=G\frac{m_1 m_2}{|\vec{r}|^3}$. Then, according to (2): $m_1\ddot{\vec{r}}_1=-\gamma\vec{r}$ and $m_2\ddot{\vec{r}}_2=\gamma\vec{r}$, and we have:

$$\frac{d}{dt}E=-\dot{\vec{r}}_1\cdot\gamma\vec{r}+\dot{\vec{r}}_2\cdot\gamma\vec{r}+\gamma\vec{r}\cdot\dot{\vec{r}}=-\dot{\vec{r}}\cdot\gamma\vec{r}+\gamma\vec{r}\cdot\dot{\vec{r}}=0$$

Thus, the following holds:

Theorem 4.2

In the motion of two bodies in each other's gravitational field, the momentum, angular momentum, and energy of the system are conserved. Furthermore, the centre of mass of the system moves with constant velocity. □

In the following, we will first choose a reference frame that is optimal for discussing these equations of motion. We note:

Lemma 4.3

Equations (2) are independent of the choice of reference frame, that is, of the choice of the origin and the orientation of the (Cartesian) coordinate system. □

Proof:

If we choose a new origin, whereby the old origin is shifted to the new one by the translation $\vec{c}$, then the following holds for the new coordinates of the two masses: $\vec{r}_{1,2}'=\vec{r}_{1,2}-\vec{c}$. It holds that $\ddot{\vec{r}}_{1,2}{}'=\ddot{\vec{r}}_{1,2}$ and $\vec{r}'=\vec{r}_1{}'-\vec{r}_2{}'=\vec{r}_1-\vec{r}_2=\vec{r}$. Thus, the systems of equations remain unchanged.

Now we rotate the coordinate system and denote this rotation by D. This rotation is bijective, and D^{-1} maps the rotated coordinate system back to the original one. Let $\vec{r}_{1,2}{}'$ be the positions of the masses in the rotated system. Then the following holds: $\vec{r}_{1,2}=D^{-1}\vec{r}_{1,2}{}'$. *D,* and thus also D^{-1}, are length-preserving, so $|\vec{r}|=|D^{-1}\vec{r}'|=|\vec{r}'|$ holds. Thus, the first equation (2) becomes:

$$D^{-1}\ddot{\vec{r}}_1'=-G\frac{m_2}{|\vec{r}'|^3}\cdot D^{-1}\vec{r}'$$

Since D^{-1} is bijective, it follows that:

$$\ddot{\vec{r}}_1' = -G\frac{m_2}{|\vec{r}'|^3}\cdot\vec{r}'$$

The same holds for the second equation. This means that the new coordinates satisfy the same equations (2) as the old ones. □

From Lemma 4.3, it follows that we can freely choose the origin of the reference frame. Here, the centre of mass suggests itself as the "natural" origin. Since it moves at a constant velocity, no external forces act in this reference frame, and we have an *inertial system*.

If the origin lies at the centre of mass, then $m_1\vec{r}_1 + m_2\vec{r}_2 = \vec{0}$ and $\vec{r}_2 = -\frac{m_1}{m_2}\vec{r}_1$. Thus, we have

$$\vec{r} = \vec{r}_1 - \vec{r}_2 = \frac{M}{m_2}\vec{r}_1$$

Now we substitute this into the first equation of Theorem 4.1 and obtain the result:

$$\ddot{\vec{r}}_1 = -G\frac{m_2\cdot\frac{M}{m_2}\vec{r}_1}{\left|\frac{M}{m_2}\vec{r}_1\right|^3} = -G\frac{m_2^3\vec{r}_1}{M^2|\vec{r}_1|^3}$$

A corresponding calculation yields an analogous result for the second equation, and we have:

Theorem 4.4

With respect to a coordinate system with its origin at the centre of mass, the equations of motion are:

$$(3)\quad \begin{cases} \ddot{\vec{r}}_1 = -G\dfrac{m_2^3}{M^2}\cdot\dfrac{\vec{r}_1}{|\vec{r}_1|^3} \\ \ddot{\vec{r}}_2 = -G\dfrac{m_1^3}{M^2}\cdot\dfrac{\vec{r}_2}{|\vec{r}_2|^3} \end{cases}$$

Here, $M = m_1 + m_2$ is the total mass.

□

It follows from Theorem 4.4 that: The two differential equations are no longer "coupled," and one can consider the trajectory of each mass relative to the centre of mass individually. We have reduced the two-body problem to a "single-body problem."

Now we can appropriately orient the coordinate system with its origin at the centre of mass. The following helps:

Lemma 4.5

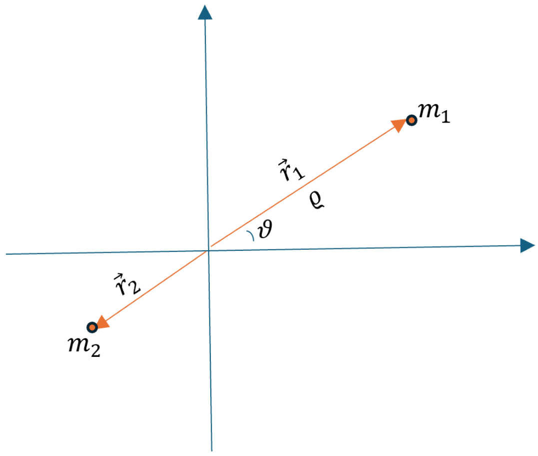
The paths of the two masses lie in the plane spanned by the difference vector $\vec{r} = \vec{r}_1 - \vec{r}_2$ and its derivative $\dot{\vec{r}}$. □

Proof:

We set $\vec{h} \coloneqq \vec{r}\times\dot{\vec{r}}$. By equations (2), we have $\ddot{\vec{r}}_{1,2} \parallel \vec{r}$, and thus $\ddot{\vec{r}} \parallel \vec{r}$. Therefore, $\dot{\vec{h}} = \vec{r}\times\ddot{\vec{r}} = \vec{0}$, so $\vec{h}$ is constant and lies perpendicular to the plane spanned by $\vec{r}$ and $\dot{\vec{r}}$. □

Thus, we can rotate the coordinate system so that the x, y plane lies in the plane spanned by $\vec{r}$ and $\dot{\vec{r}}$. By Lemma 4.5, the motion occurs in this plane. By Lemma 4.3, we still have the same equations of motion in the new coordinates, and we omit marking the new coordinates with an ‘.

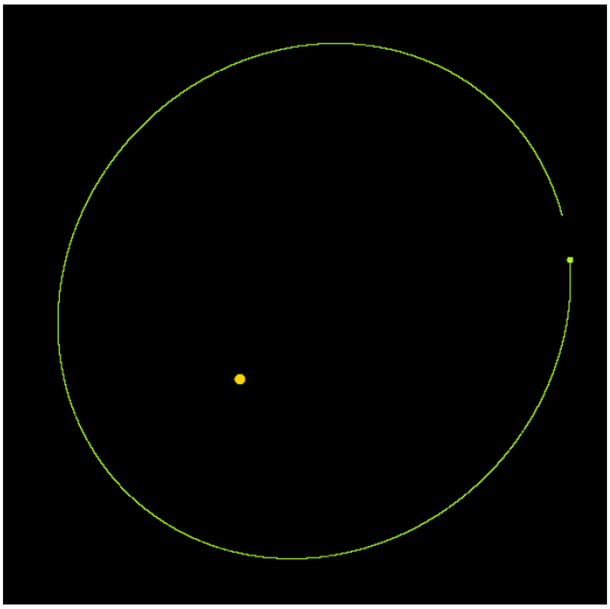
The following sketch illustrates the corresponding situation.



Position of the masses relative to the coordinate system at the centre of mass and the $\vec{r}, \dot{\vec{r}}$ plane

If the motion now takes place in the x, y plane, the equations (3)—decomposed into components—represent two second-order differential equations.

In the simulator, these equations are numerically approximated using the Runge-Kutta method. This method was explained in the school project *“Numerical Methods and Coupled* Pendulums.” In the simulator, you can change the starting position, the mass, and also the velocity (at perihelion and thus perpendicular to the position vector) of a planet and experiment with these parameters.



Simulation of Jupiter’s orbit, starting from a slightly shifted position

5. Solution to the two-body problem

In the following, we seek a solution to the first equation (3) from Theorem 4.4 for the mass m_1 and need only consider the x, y plane.

We first introduce *polar coordinates* (see previous sketch):

$$\vec{r}_1 = \varrho \begin{pmatrix} cos\vartheta \\ sin\vartheta \end{pmatrix}, |\vec{r}_1| = \varrho$$

$$\dot{\vec{r}}_1 = \dot{\varrho} \begin{pmatrix} cos\vartheta \\ sin\vartheta \end{pmatrix} + \varrho\dot{\vartheta} \begin{pmatrix} -sin\vartheta \\ cos\vartheta \end{pmatrix}$$

$$\ddot{\vec{r}}_1 = \ddot{\varrho} \begin{pmatrix} cos\vartheta \\ sin\vartheta \end{pmatrix} + 2\dot{\varrho}\dot{\vartheta} \begin{pmatrix} -sin\vartheta \\ cos\vartheta \end{pmatrix} + \varrho\ddot{\vartheta} \begin{pmatrix} -sin\vartheta \\ cos\vartheta \end{pmatrix} + \varrho\dot{\vartheta}^2 \begin{pmatrix} -cos\vartheta \\ -sin\vartheta \end{pmatrix}$$

The goal is now to find two equations for $\varrho(t)$ and $\vartheta(t)$. Then, we eliminate t from these equations so that we obtain an equation for $\varrho(\vartheta)$ and thus the desired equation of motion. To do this, we consider the angular momentum $\vec{L}$ and the energy E of the mass m_1. We use the fact that the angular momentum is constant:

$$\dot{\vec{L}} := \frac{d}{dt}(\vec{r}_1 \times m_1\dot{\vec{r}}_1) = \vec{r}_1 \times m_1\ddot{\vec{r}}_1 = \vec{0}$$

since according to (3): $\ddot{\vec{r}}_1 \parallel \vec{r}_1$. The angular momentum is perpendicular to the coordinate plane, and we consider only its magnitude. Then, in polar coordinates:

$$L := |\vec{L}| = |\vec{r}_1 \times m_1\dot{\vec{r}}_1| = m_1\varrho^2\dot{\vartheta} = \text{constant}$$

This allows us to eliminate the derivatives with respect to time.

First, we obtain:

$$\dot{\vartheta} = \frac{L}{m_1} \cdot \frac{1}{\varrho^2}$$

Furthermore:

$$\dot{\varrho} = \frac{d\varrho}{dt} = \frac{d\varrho}{d\vartheta} \cdot \frac{d\vartheta}{dt} = \varrho'\dot{\vartheta} = \frac{L}{m_1} \cdot \frac{\varrho'}{\varrho^2}$$

Where we set: $\varrho' = \frac{d\varrho}{d\vartheta}$. Now let's look again at equation (3):

$$m_1\ddot{\vec{r}}_1 = -G\frac{m_2^3 m_1}{M^2} \cdot \frac{\vec{r}_1}{|\vec{r}_1|^3}$$

A force of $-G\frac{m_2^3 m_1}{M^2} \cdot \frac{\vec{r}_1}{|\vec{r}_1|^3}$ acts on m_1, and the corresponding potential energy is (if the origin is chosen at infinity):

$$E_{pot} = -G\frac{m_2^3 m_1}{M^2} \cdot \frac{1}{|\vec{r}_1|}$$

Thus, the energy of the mass m_1 can be written as:

$$E = \frac{1}{2}m_1|\dot{\vec{r}}_1|^2 - G\frac{m_2^3 m_1}{M^2} \cdot \frac{1}{|\vec{r}_1|}$$

In polar coordinates, we have:

$$E = \frac{m_1}{2}\left(\dot{\varrho}^2 + \varrho^2\dot{\vartheta}^2\right) - G\frac{m_2^3 m_1}{M^2}\cdot\frac{1}{\varrho}$$

Now we eliminate $\dot{\varrho}, \dot{\vartheta}$:

$$E = \frac{m_1}{2}[\left(\frac{L}{m_1}\right)^2 \cdot \frac{\varrho'^2}{\varrho^4} + \varrho^2\left(\frac{L}{m_1}\right)^2 \cdot \frac{1}{\varrho^4}] - G\frac{m_2^3 m_1}{M^2}\cdot\frac{1}{\varrho}$$

And we obtain:

$$E = \frac{L^2}{2m_1}\cdot\frac{1}{\varrho^2}\left(\frac{\varrho'^2}{\varrho^2} + 1\right) - G\frac{m_2^3 m_1}{M^2}\cdot\frac{1}{\varrho}$$

If the mass m_1 flies directly toward the centre of mass and thus the other mass m_2, then $\vec{r}_1 \parallel \dot{\vec{r}}_1$ and thus $L = 0$. Then we simply have a rectilinear motion that ends with the collision of the masses. We exclude this case. Thus, in what follows, $L \neq 0$ and we divide by L. This gives us a first-order differential equation in which time$t$ is eliminated:

Lemma 5.1

In polar coordinates, the motion of the massm_1 and the desired function $\varrho(\vartheta)$ satisfy the first-order differential equation:

$$(4) \quad \frac{2m_1 E}{L^2} = \frac{1}{\varrho^2}\left(\frac{\varrho'^2}{\varrho^2} + 1\right) - 2G\frac{m_2^3 m_1^2}{M^2 L^2}\cdot\frac{1}{\varrho}$$

□

We now seek a solution to the differential equation (4).

We set: $p := \frac{M^2 L^2}{G m_2^3 m_1^2}$ and substitute: $\varrho = \frac{1}{u}$, $\varrho' = -\frac{u'}{u^2}$. This gives us:

$$\frac{2m_1 E}{L^2} = u'^2 + u^2 - \frac{2u}{p}$$

Completing the square yields:

$$\frac{2m_1 E}{L^2} = u'^2 + (u - \frac{1}{p})^2 - \frac{1}{p^2}$$

We make the substitution:

$$u(\vartheta) = A\cos(\vartheta - \alpha) + \frac{1}{p} \ , u'(\vartheta) = -A\sin(\vartheta - \alpha)$$

and by substituting, we obtain:

$$\frac{2m_1 E}{L^2} = A^2 - \frac{1}{p^2}$$

Then we obtain:

$$A^2 = \frac{2m_1 E}{L^2} + \frac{1}{p^2}$$

First, we want to check whether this expression is always defined, that is, whether the right-hand side is always positive.

E is constant. If we consider the energy at perihelion or aphelion, there $\dot{\varrho} = 0$ and we obtain for the kinetic energy at this point:

$$E_{kin} = \frac{m_1}{2}\varrho^2\dot{\vartheta}^2 = \frac{m_1}{2}\varrho^2\frac{L^2}{m_1^2\varrho^4} = \frac{L^2}{2m_1\varrho^2}$$

And for the total energy:

$$E = \frac{L^2}{2m_1\varrho^2} - G\frac{m_2^3 m_1}{M^2}\cdot\frac{1}{\varrho} = \frac{L^2}{2m_1}\left(\frac{1}{\varrho^2} - \frac{2}{p}\cdot\frac{1}{\varrho}\right)$$

Thus:

$$\frac{2m_1 E}{L^2} + \frac{1}{p^2} = \frac{1}{\varrho^2} - \frac{2}{p}\cdot\frac{1}{\varrho} + \frac{1}{p^2} = \left(\frac{1}{\varrho} - \frac{1}{p}\right)^2 \geq 0$$

□

Now:

$$A^2 = \frac{2m_1 E}{L^2} + \frac{1}{p^2} = \frac{1}{p^2}\left(\frac{2m_1 E}{L^2}\cdot p^2 + 1\right) = \frac{1}{p^2}\cdot\left(\frac{2EL^2M^4}{G^2m_1^3m_2^6} + 1\right)$$

We defineε as:

$$\varepsilon^2 := A^2p^2 = \frac{2EL^2M^4}{G^2m_1^3m_2^6} + 1 \geq 0$$

It suffices to consider the case$\varepsilon \geq 0$; otherwise, one can simply add π to $\vartheta - \alpha$.

We have: $A = {}^{\varepsilon}\!/_{p}$ and

$$u(\vartheta) = \frac{\varepsilon\cos(\vartheta - \alpha) + 1}{p}$$

And thus:

Theorem 5.2

The differential equation (4) has the solution:

$$(5)\ \ \varrho(\vartheta) = \frac{p}{1 + \varepsilon\cos(\vartheta - \alpha)}$$

□

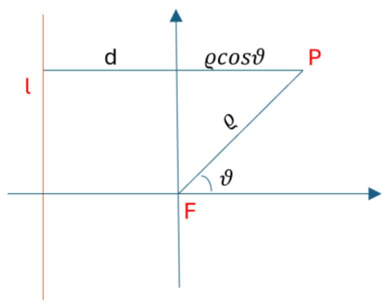
Theorem 5.3

The path of the mass m_1 around the centre of mass is a *conic section*.

□

Proof

A conic section can be defined as the set of points that maintain a constant ratio of distancesε from a fixed point (the focus F) and a fixed line (the directrix l):



The point P and its distance from F or l

$$\frac{\varrho}{d + \varrho\cos\vartheta} = \varepsilon$$

$$\varrho = \frac{p}{1 - \varepsilon\cos\vartheta}$$

$p = \varepsilon d = \varrho(\frac{\pi}{2})$ is half the width of the conic section at the focus. For $\alpha = \pi$, equation (5) agrees with the above equation.

The minimum distance of a planet from the Sun or from the focus is the *perihelion*. If we look at equation (5) again:

$$\varrho(\vartheta) = \frac{p}{1 + \varepsilon\cos{(\vartheta - \alpha)}}$$

Then we see that the minimum is assumed for $\cos(\vartheta - \alpha) = 1$ or $\vartheta = \alpha$.

α is precisely *the argument of the perihelion*, or the angle between the direction of the perihelion and the$x$ axis. In our solar system, the$x$ axis is defined by the intersection of the ecliptic and the Earth's equatorial plane at the vernal equinox.

Depending on the value of the numerical eccentricity ε, one obtains the various types of conic sections:

$\varepsilon = 0$: $\varrho = p = \text{constant}$. The conic section is a *circle*

$0 < \varepsilon < 1$:$\frac{p}{1-\varepsilon} < \varrho < \frac{p}{1+\varepsilon}$. The conic section is an *ellipse*.

$\varepsilon = 1$: $\varrho \to \infty$ for$\vartheta \to 0$. Otherwise, $\varrho > 0$ and defined for all $\vartheta \in]0,2\pi[$. The result is a *parabola*.

$\varepsilon > 1$: $\varrho > 0$ only for $\cos\vartheta < \frac{1}{\varepsilon}$. $\varrho \to \infty$ for $\vartheta \to \arccos\frac{1}{\varepsilon}$. We have a *hyperbola*.

□

Note

We performed our calculation with the centre of mass as the origin. In the literature, the location of the mass m_2 is sometimes set as the origin, and the motion of m_1 is calculated relative to m_2. In that case, slightly different values are obtained for the parameters p, ε in the equation of motion.

Finally, let us consider the units of the parameters that appeared in the equation of motion for the two-body problem. This was:

$$\varrho(\vartheta) = \frac{p}{1 + \varepsilon\cos{(\vartheta - \alpha)}}$$

In it, we had: `$p := \frac{M^2L^2}{Gm_1^2m_2^3}$`. To denote the unit of a quantity, we write that quantity in square brackets and use the kilogram–meter–second system. Then, `$[L] = \frac{kg \cdot m^2}{s}$`. Thus:

$$[p] = [M]^2[L]^2 \cdot [Gm_1^2m_2^3]^{-1} = kg^2 \cdot \frac{kg^2 \cdot m^4}{s^2} \cdot \frac{kg \cdot s^2}{m^3} \cdot \frac{1}{kg^5} = m$$

p thus has the dimension of a length.

Furthermore:

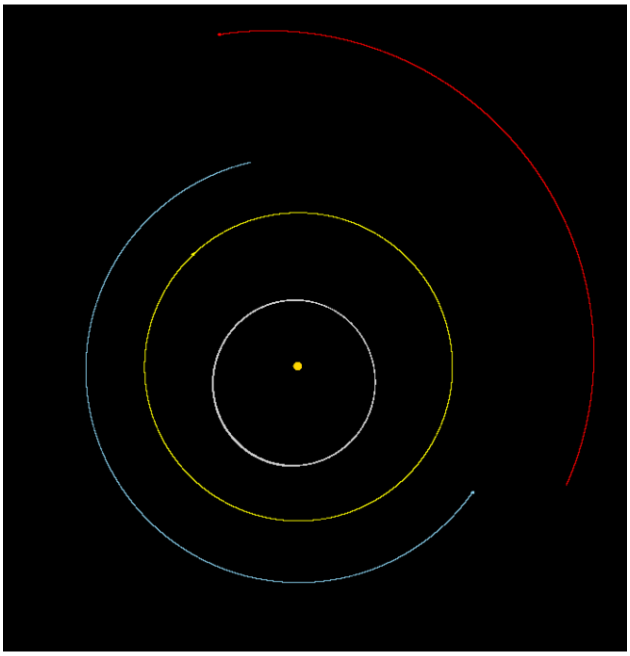
$$\varepsilon^2 := \frac{2EL^2M^4}{G^2m_1^3m_2^6} + 1$$

It is:

$$\left[\frac{EL^2M^4}{G^2m_1^3m_2^6}\right] = [EL^2M^4] \cdot [G^2m_1^3m_2^6]^{-1} = \frac{kg \cdot m^2}{s^2} \cdot \frac{kg^2 \cdot m^4}{s^2} \cdot kg^4 \cdot \frac{kg^2 \cdot s^4}{m^6} \cdot \frac{1}{kg^9} = 1$$

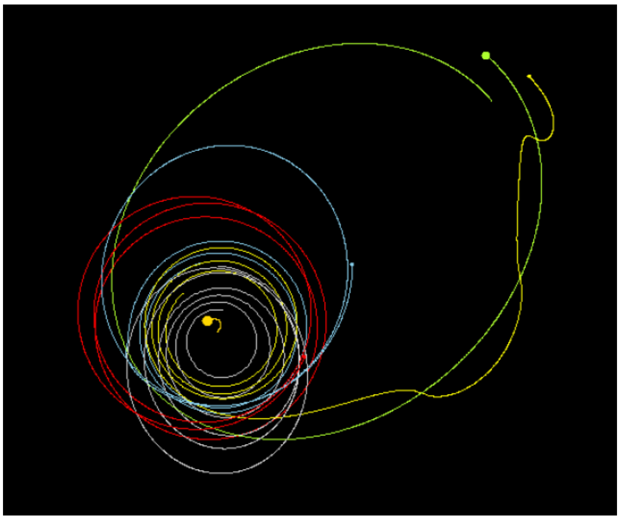
ε is dimensionless.

What does this look like in our planetary system? The mass of the Sun is about 750 times greater than the total mass of all the planets combined. Suppose all the planets were aligned in a straight line. Even then, the common centre of mass of the entire solar system would be only about $\frac{1}{100}$ AE outside the centre of the Sun (this can be verified in an exercise).1 AE is the *astronomical unit*, i.e., the average distance between Earth and the Sun (see Appendix). For Kepler, this meant that the Sun was at the focus of an elliptical planetary orbit. The masses of the planets are so small compared to their mutual distances that they hardly interact. One can therefore consider the planetary orbits as individual, isolated two-body problems. Below is a simulation in the simulator with Mercury, Venus, Earth, Mars, and the Sun at the centre:



Simulated orbits of the inner planets around the Sun during a Venusian year (224.7 days)

Suppose a "vagabond" planet, ejected from another solar system, were to pass through our solar system. If it were large enough, its gravitational force would affect our planets and possibly even knock them out of their orbits. This would spell the end of the stability of the planetary orbits. Such scenarios can be simulated in the simulator. In the following, such a rogue planet encounters the inner planets.



A “vagabond” (green) with 20 times the mass of Jupiter has knocked Earth (blue) and Mercury (gray) out of their orbits and “captured” Venus (yellow). Mars’s orbit (red) also becomes unstable. In addition, the vagabond disrupts the Sun’s stability.

6. The Law of Gravitation and Kepler’s Laws

Newton’s achievement was not only to correctly formulate the law of gravitation, but above all to prove that Kepler’s laws follow from it. We will derive this as well.

Theorem 5.3 directly yields Kepler’s first law: *The planets move in ellipses with the Sun at their focus*.

An infinitesimal area element swept out by the position vector is, in polar coordinates:

$$dF = \frac{1}{2}\varrho^2\dot{\vartheta}dt = \frac{1}{2}\frac{L}{m_1}dt$$

since the angular momentum L is constant. Thus, the area swept out during a fixed time Δt is always the same. From this follows Kepler’s second law: *The line connecting a planet to the Sun sweeps out equal areas in equal times*.

If T is the orbital period of the mass m_1, then the area of the ellipse $F = \pi ab$ is swept out during this time. Therefore, the following holds:

$$\pi ab = \int_0^T \frac{1}{2}\varrho^2\dot{\vartheta}dt = \frac{1}{2}\int_0^T \frac{L}{m_1}dt = \frac{1}{2}T\frac{L}{m_1}$$

Thus:

$$T = 2\pi ab\frac{m_1}{L}$$

The definition of *p* was: $p := \frac{M^2L^2}{Gm_2^3m_1^2}$. Furthermore, $p = a(1-\varepsilon^2)$, and thus

$$L = \sqrt{Gm_2} \cdot \frac{m_1 m_2}{M} \sqrt{a(1-\varepsilon^2)}$$

Furthermore, $b = a\sqrt{1-\varepsilon^2}$. If we substitute this into the above formula for T, we obtain:

$$T = 2\pi a^2 \sqrt{1-\varepsilon^2} \cdot m_1 \frac{M}{\sqrt{Gm_2} m_1 m_2 \sqrt{a(1-\varepsilon^2)}} = 2\pi \sqrt{a}^3 \frac{M}{\sqrt{Gm_2^3}}$$

Thus:

$$\frac{T^2}{a^3} = 4\pi^2 \frac{M^2}{Gm_2^3} = \text{constant}$$

In the special case of the Sun and a planet, $m_2 = M_{Sun}$ and $M = m_1 + m_2 \approx M_{Sun}$

Then:

$$\frac{T^2}{a^3} = 4\pi^2 \frac{1}{GM_{Sun}}$$

From this follows Kepler's third law: *The squares of the orbital periods are in the same ratio as the cubes of the semi-major axes*.

As a result, we have

Theorem 6.1

Kepler's laws follow from Newton's law of gravitation. □

7. The 3-Body Problem

Since the two-body problem had a closed-form analytical solution, Newton began investigating how three bodies move in each other's gravitational field. We will investigate this as well. We denote the three masses by m_1, m_2, m_3 and their positions by $\vec{r}_1, \vec{r}_2, \vec{r}_3$.

Then the equations of motion are:

$$(6) \begin{cases} m_1 \ddot{\vec{r}}_1 = G \frac{m_1 m_2}{|\vec{r}_2 - \vec{r}_1|^3} (\vec{r}_2 - \vec{r}_1) + G \frac{m_1 m_3}{|\vec{r}_3 - \vec{r}_1|^3} (\vec{r}_3 - \vec{r}_1) \\ m_2 \ddot{\vec{r}}_2 = G \frac{m_2 m_3}{|\vec{r}_3 - \vec{r}_2|^3} (\vec{r}_3 - \vec{r}_2) + G \frac{m_2 m_1}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2) \\ m_3 \ddot{\vec{r}}_3 = G \frac{m_3 m_1}{|\vec{r}_1 - \vec{r}_3|^3} (\vec{r}_1 - \vec{r}_3) + G \frac{m_3 m_2}{|\vec{r}_2 - \vec{r}_3|^3} (\vec{r}_2 - \vec{r}_3) \end{cases}$$

As can be seen, the system is *symmetric*. If the indices$1 \to 2, 2 \to 3, 3 \to 1$ are cyclically permuted, one equation merges into the next. We can also write this compactly:

$$(7)\ \ddot{\vec{r}}_i = G \sum_{i \neq j} \frac{m_j}{\left|\vec{r}_j - \vec{r}_i\right|^3} \left(\vec{r}_j - \vec{r}_i\right)\ , i = 1,2,3$$

If the origin of the coordinate system is shifted by the vector $\vec{c}$, the equations do not change. As with the two-body problem, the same applies if the coordinate system is rotated by a rotation D. Thus, the three-body problem is *independent of the choice of origin and the orientation of the (Cartesian) coordinate system.*

The centre of mass of the system with coordinates $\vec{r}_s$ is defined by:

$$M\vec{r}_s = m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3$$

Where $M = m_1 + m_2 + m_3$.

If we add the equations (1), the right-hand side cancels out because the terms appear in pairs with opposite signs:

$$m_1\ddot{\vec{r}}_1 + m_2\ddot{\vec{r}}_2 + m_3\ddot{\vec{r}}_3 = \vec{0}$$

This leads to the *law of conservation of momentum*:

$$m_1\dot{\vec{r}}_1 + m_2\dot{\vec{r}}_2 + m_3\dot{\vec{r}}_3 = \vec{p} = \text{constant}$$

Thus, $M\dot{\vec{r}}_s = \text{constant}$ also holds, and the centre of mass moves at a constant velocity. We can therefore choose the centre of mass as the origin of the coordinate system and obtain an *inertial system*.

For the angular momentum $\vec{L}$, the following applies:

$$\frac{d}{dt}\vec{L} = \frac{d}{dt}\sum_{j=1}^{3}\vec{r}_j \times m_j\dot{\vec{r}}_j = \sum_{j=1}^{3}\vec{r}_j \times m_j\ddot{\vec{r}}_j = \sum_{j=1}^{3}\vec{r}_j \times \{\sum_{i\neq j} G\frac{m_j m_i}{\left|\vec{r}_i - \vec{r}_j\right|^3}\cdot(\vec{r}_i - \vec{r}_j)\}$$

In this sum, the terms always appear in pairs, e.g.,

$\vec{r}_1 \times G\frac{m_1 m_2}{|\vec{r}_2-\vec{r}_1|^3}\cdot(\vec{r}_2 - \vec{r}_1) = \vec{r}_1 \times G\frac{m_1 m_2}{|\vec{r}_2-\vec{r}_1|^3}\cdot\vec{r}_2$ and$\vec{r}_2 \times G\frac{m_1 m_2}{|\vec{r}_1-\vec{r}_2|^3}\cdot(\vec{r}_1 - \vec{r}_2) = \vec{r}_2 \times G\frac{m_1 m_2}{|\vec{r}_2-\vec{r}_1|^3}\cdot\vec{r}_1$. The sum of such a pair is zero, so the total is $\frac{d}{dt}\vec{L} = 0$, and thus the *angular momentum* is *constant*.

When examining the total energy of the system, we first need the potential energy. To do this, we consider the potential energy of the mass m_2 in the gravitational field of the mass m_1. To this we add the potential energy of the mass m_3 in the gravitational fields of m_1 and m_2. This gives us, as in the two-body problem:

$$E_{pot} = G\frac{m_1 m_2}{|\vec{r}_2 - \vec{r}_1|} + G\frac{m_1 m_3}{|\vec{r}_3 - \vec{r}_1|} + G\frac{m_2 m_3}{|\vec{r}_3 - \vec{r}_2|}$$

The kinetic energy is:

$$E_{kin} = \frac{1}{2}m_1\left|\dot{\vec{r}}_1\right|^2 + \frac{1}{2}m_2\left|\dot{\vec{r}}_2\right|^2 + \frac{1}{2}m_3\left|\dot{\vec{r}}_3\right|^2$$

A calculation that is not difficult but somewhat involved then yields, using equations (6):

$$\frac{d}{dt}\left(E_{kin} + E_{pot}\right) = 0$$

The *total energy is therefore constant,* and the following holds:

Theorem 7.1

In the motion of three bodies in each other's gravitational field, the momentum, angular momentum, and energy of the system are conserved. Furthermore, the centre of mass of the system moves with constant velocity. □

Since the angular momentum $\vec{L}$ is constant, the plane perpendicular to $\vec{L}$ is invariant. It is called the *invariant plane of Laplace*. We can orient the coordinate system such that the plane x, y coincides with this invariant plane. If the velocity vectors of the three masses at the start all lie in the plane passing through the three masses, *this plane remains invariant and the motion proceeds in this plane.*

A detailed presentation of the mathematical theory of the N-body problem can be found in “Orbital Motion” [1] and “Celestial Mechanics” [2]. Both books also introduce celestial mechanics while remaining at an accessible mathematical level.

8. Lagrange’s Parking Spots

Newton investigated the three-body problem without arriving at an analytical solution. Subsequently, virtually all prominent mathematicians have struggled with this problem. When we observe a simulation in the simulator, it is plausible that there is likely no analytical solution to the three-body problem.

The orbits of three masses of equal weight in each other’s gravitational field

In this simulation, three masses of equal size were launched from the points $(-1,0)$, $(0,0)$, $(1,0)$ with different initial velocities.

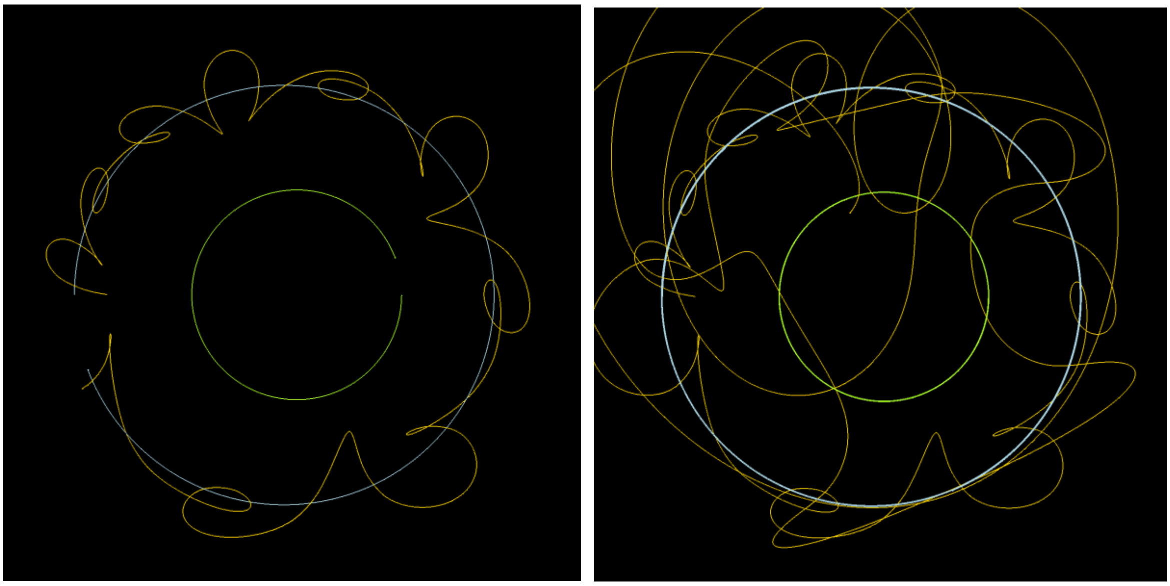
When a problem seems unsolvable, one can at least try to consider simple special cases. Leonhard Euler (1707–1783) investigated the case in which all three bodies move in a straight line. Joseph-Louis Lagrange (1736–1813), whom we already encountered in the school project *“Numerical*

Methods and Coupled Pendulums," investigated the following special case of the three-body problem, known as *the restricted three-body problem*:

Definition 8.1 (*Constrained Three-Body Problem*)

1) Two masses M_1 andM_2 rotate around their common centre of mass. Since M_1 and M_2 move as in the two-body problem, their trajectories lie in a common plane.
2) A third body with mass m moves in the same plane as the first two masses and within their gravitational field. However, its mass is so small compared to M_1 andM_2 that its gravitational force in the system is negligible.

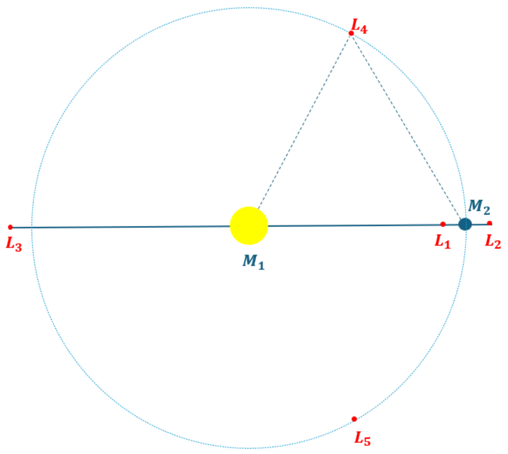
Below is a simulation in the simulator, in which the third body also lies in this plane. Its trajectory is shown in yellow.



At first, the yellow body "wobbles" around the blue one. Later, it also approaches the green one.

Evidently, this special case also leads to complicated orbits for the third body. However, Lagrange found five equilibrium points for the third body, such that it does not change its position relative to the other bodies when the system as a whole rotates around the centre of mass. These Lagrange points lie in the orbital plane of the two large masses. Lagrange's solution to this special case—which was hypothetical in his time—is of great importance today for "parking" satellites in the Earth-Moon system.

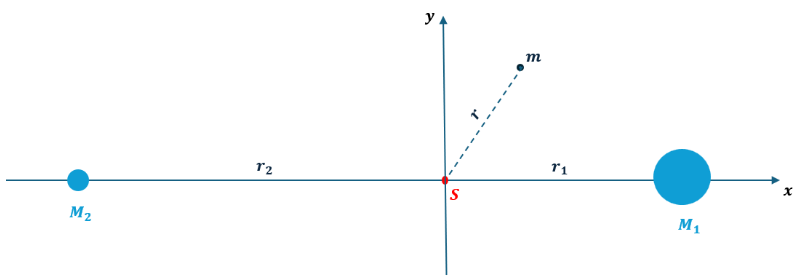
Below is a sketch showing the Lagrange points. In this diagram, $M_1 \gg M_2$, so that the common centre of mass lies within M_1, this mass moves very little, and M_2 orbits M_1. The Lagrange points are marked in red. When the mass m is located there, the forces acting on it (gravitational forces and centrifugal force) cancel each other out. This mass can be "parked" there.



Sketch with Lagrange points

The first three Lagrange points $L_{1,2,3}$ are unstable in the direction of the line $M_1 - M_2$ and stable perpendicular to it. They form homocline equilibrium points, as we learned in the school project *"Strange Attractors and Edward Lorenz's Weather Forecast*." In contrast, L_4 and L_5 are stable, provided that approximately $M_1 > 25 \cdot M_2$. M_1, M_2, L_4 and M_1, M_2, L_5 form an equilateral triangle.

To analyse this problem, let's consider an even simpler special case: We assume that the two heavy masses rotate in *circular paths* around the centre of mass S. This means that their distance from the centre of mass remains constant. The centre of mass lies on the line connecting the two masses. If we choose it as the origin of the coordinate system, then this line rotates around the centre of mass at a constant angular velocity. Now we choose this line as the x-axis of the coordinate system and the y-axis perpendicular to it. This coordinate system rotates around the centre of mass, so it is not an inertial system. However, the massive bodies are at rest in this system, and all points on the x-axis remain fixed.



The three bodies in our rotating coordinate system

The condition for the centre of mass is:

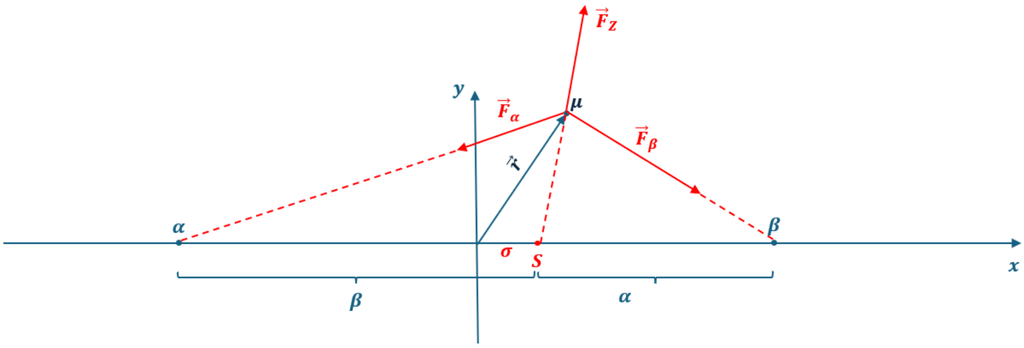
$$M_1 r_1 = M_2 r_2$$

In addition to gravitational forces, the mass m is also subject to centrifugal force. In Theorem 3.1, we established according to Huygens that the following applies to this:

$$F_z = m\frac{v^2}{r} = mr\omega^2$$

Where v is the tangential velocity perpendicular to r and ω is the angular velocity.

To investigate this system, we consider the following abstract mathematical model:



The abstract mathematical model

The two massive objects are represented by two points on the x-axis, to which the numerical values $\alpha > 0, \beta > 0$ are assigned, representing the magnitude of these masses. We can choose the unit of mass such that the following holds: $\alpha + \beta = 2$, where we do not explicitly specify the physical unit of mass. We choose the unit of length for the x-axis such that the distance between the points α, β is exactly 2. Since we have chosen the midpoint of the line segment $\overline{\alpha\beta}$ as the origin, these points have the coordinates $(\pm 1{,}0)$. We again denote the centre of mass of the masses α, β by S. The condition for the centre of mass is satisfied if α has the distance β from S and β has the distance α from S. Then the following holds: $\sigma = 1 - \alpha = \beta - 1$; by definition, we can additionally assume that $\beta \geq \alpha$. Then $0 \leq \sigma < 1$ holds as outlined.

Now let us consider the mass μ (which is tiny compared to α, β) at the position $\vec{r} = \binom{x}{y}$. This mass is subject to the gravitational forces $\vec{F}_\alpha$ and $\vec{F}_\beta$, as well as the centrifugal force $\vec{F}_Z$, which arises from the rotation of the coordinate system. We determine its angular velocity ω using the masses α, β, which are at rest in this coordinate system.

α is subject to the centrifugal force $\alpha\beta\omega^2$, which exactly compensates for the gravitational force of β. So far, we have chosen appropriate units for mass and length. Now we choose the unit of time such that the gravitational constant in the new units is $\tilde{G} = 1$. Then the following holds:

$$\alpha\beta\omega^2 = \tilde{G}\frac{\alpha\beta}{(\alpha+\beta)^2} = \frac{\alpha\beta}{4}$$

Thus, $\omega^2 = \frac{1}{4}$. We obtain the same result when we consider the forces acting on the mass β.

What forces act on the mass μ? The direction of $\vec{F}_\alpha$ is $\binom{-1-x}{-y}$ and that of $\vec{F}_\beta$ is $\binom{1-x}{-y}$. According to Newton's law of gravitation, we then have:

$$\vec{F}_\alpha = \frac{\alpha\mu}{\sqrt{(-1-x)^2+(-y)^2}^3}\begin{pmatrix}-1-x\\ -y\end{pmatrix}$$

$$\vec{F}_\beta = \frac{\beta\mu}{\sqrt{(1-x)^2+(-y)^2}^3}\begin{pmatrix}1-x\\ -y\end{pmatrix}$$

The centrifugal force acts in the direction $\vec{r} - \binom{\sigma}{0} = \binom{x-1+\alpha}{y}$ and the following applies:

$$\vec{F}_Z = \mu\sqrt{(x+\alpha-1)^2+y^2}\,\omega^2\frac{\binom{x+\alpha-1}{y}}{\sqrt{(x+\alpha-1)^2+y^2}} = \frac{\mu}{4}\begin{pmatrix}x+\alpha-1\\ y\end{pmatrix}$$

Now we can write down the law of motion for the mass μ. According to Newton, and considering the forces acting on it, the following holds:

$$\mu\ddot{\vec{r}} = \frac{-\alpha\mu}{\sqrt{(1+x)^2+y^2}^3}\begin{pmatrix}1+x\\ y\end{pmatrix} + \frac{\beta\mu}{\sqrt{(1-x)^2+y^2}^3}\begin{pmatrix}1-x\\ -y\end{pmatrix} + \frac{\mu}{4}\begin{pmatrix}x+\alpha-1\\ y\end{pmatrix}$$

The parameters α, β are interdependent, and we substitute: $\alpha = 1-\sigma$ and $\beta = 1+\sigma$, so that we are left with only one parameter σ. Result:

Lemma 8.2

Under the previous notations and assumptions, the following law of motion applies to the mass μ:

$$(7)\ \begin{cases} \ddot{x} = \dfrac{-(1-\sigma)(1+x)}{\sqrt{(1+x)^2+y^2}^3} + \dfrac{(1+\sigma)(1-x)}{\sqrt{(1-x)^2+y^2}^3} + \dfrac{x-\sigma}{4} \\ \ddot{y} = \dfrac{-(1-\sigma)y}{\sqrt{(1+x)^2+y^2}^3} + \dfrac{-(1+\sigma)y}{\sqrt{(1-x)^2+y^2}^3} + \dfrac{y}{4} \end{cases}$$

Here, the following holds for the parameter $0 \leq \sigma < 1$.

□

For the equilibrium positions of the mass μ, $\ddot{x} = 0 = \ddot{y}$ holds, and thus:

$$(8)\ \begin{cases} \dfrac{-(1-\sigma)(1+x)}{\sqrt{(1+x)^2+y^2}^3} + \dfrac{(1+\sigma)(1-x)}{\sqrt{(1-x)^2+y^2}^3} = \dfrac{\sigma-x}{4} \\ \dfrac{(1-\sigma)y}{\sqrt{(1+x)^2+y^2}^3} + \dfrac{(1+\sigma)y}{\sqrt{(1-x)^2+y^2}^3} = \dfrac{y}{4} \end{cases}$$

Clearly, the equilibrium positions are symmetric about the x-axis. If we replace y with $-y$, the equations remain unchanged.

First, we show that L_4 and L_5 are equilibrium positions. Since these points form an equilateral triangle with α, β, their coordinates satisfy $L_{4,5} = (0, \pm\sqrt{3})$. For these values of x, y, we have:

$$\sqrt{(1+x)^2+y^2} = \sqrt{(1-x)^2+y^2} = 2$$

And it is easy to verify that both equations are satisfied.

Showing that these are the only solutions with $y \neq 0$ is physically plausible, but mathematically difficult to prove, so we will omit this. We will also consider the case $y = 0$. These are the equilibrium positions on the x-axis. The second equation is then satisfied, and the first takes the form:

$$-\frac{(1-\sigma)(1+x)}{\sqrt{(1+x)^2}^3} + \frac{(1+\sigma)(1-x)}{\sqrt{(1-x)^2}^3} = \frac{\sigma-x}{4}$$

The fractions on the left-hand side can be simplified, but one must consider the sign of the numerator, since the denominator contains the absolute value of the numerator. Depending on the sign, one then has an equation of the form:

$$\mp\frac{1-\sigma}{(1+x)^2} \pm \frac{1+\sigma}{(1-x)^2} = \frac{\sigma-x}{4}$$

This yields the following cases:

1) $1 < x$: Then $1 + x > 0, 1 - x < 0$, and the equation has “-”, “-”. If a solution exists, this yields the equilibrium position L_2.
2) $-1 < x < 1$: Then $1 + x > 0, 1 - x > 0$, and the equation has “-”, “+”. If a solution exists, this yields the equilibrium position L_1.
3) $x < -1$. Then $1 + x < 0, 1 - x > 0$, and the equation yields “+”, “+”. If a solution exists, this gives the equilibrium position L_3.
4) The case "+", "-" would imply: $1 + x < 0$ and $1 - x < 0$. That is not possible.

To verify that solutions actually exist in all three cases, let’s examine case 2) as an example. On the right-hand side of the equation, we have a line passing through the points $(0, \frac{\sigma}{4})$ and $(\sigma, 0)$. The left-hand side of the equation is a function

$$f(x) = -\frac{1-\sigma}{(1+x)^2} + \frac{1+\sigma}{(1-x)^2}, x \in]-1,1[$$

It is $\lim_{x \to -1^+} f(x) = -\infty$ and $\lim_{x \to 1^-} f(x) = \infty$. Between these points, the function is continuous and intersects the line on the right-hand side of the equation.

To summarize:

Theorem 8.3 (*Lagrange’s parking lots*)

For the equilibrium positions of the mathematical model outlined in this section, the following holds:

a) The equilibrium points $L_{1,2,3}$ on the x-axis are solutions to the equation:
$$\mp \frac{1-\sigma}{(1+x)^2} \pm \frac{1+\sigma}{(1-x)^2} = \frac{\sigma - x}{4}$$
Where $0 \leq \sigma < 1$.
b) $L_{4,5} = (0, \pm\sqrt{3})$ are symmetric equilibrium positions outside the x-axis. □

9. Henri Poincaré and the King of Sweden

If one considers the differential equations (6) of the three-body problem, only one additional equation is added compared to the two-body problem. The symmetry between the equations is preserved, and at first glance it does not seem impossible that this problem is also solvable. In our planetary system, the Sun–Earth–Moon system is a three-body system in which these bodies influence one another, though in the case of the Sun, this influence acts in only one direction. Nevertheless, this system results in periodic and stable orbits, at least according to our experience. Therefore, these orbits should be derivable from the differential equations (6). However, all attempts to solve system (6) in general have failed. About 100 years after Newton, Lagrange’s points represented arguably the greatest advance in the study of the problem. This already hinted at its complexity.

King Oscar II of Sweden (1829–1907) was very interested in culture and science. He was an honorary member of the Royal Swedish Academy of Sciences. In 1885, Oscar II offered a prize for the solution to the N-body problem, with the aim, among other things, of proving the stability of the planetary system. The prize money at the time was 2,500 Swedish kronor—which would be about 18,000 CHF today—plus a gold medal. That may seem modest, but at the time it was roughly five times the annual salary of an ordinary worker.

All attempts at the time to solve the problem relied on purely computational methods. In middle school, students are introduced to various differential equations, mostly only those that can be solved using various, sometimes ingenious, tricks. In the 18th and 19th centuries, many mathematicians—some of them great—worked on solving such equations.

The decisive breakthrough in the three-body problem was achieved by Henri Poincaré (1854–1912). He was one of the most significant and versatile mathematicians of his time and made substantial contributions to all areas of mathematics. Henri Poincaré eschewed computational tricks to solve the three-body problem and instead considered the geometric aspects of the problem. In doing so, he discovered its unsolvability and its high complexity, which, from today's perspective, proves to be chaotic. With his work, he was one of the founders of modern chaos theory.

Henri Poincaré was such a versatile and gifted mathematician that he tackled the three-body problem not so much for the prize money on offer, but rather out of pure mathematical interest. He did, however, ultimately receive the prize. Not because he was able to solve the problem analytically, but because his work offered a much deeper insight into the problem. It became clear that this problem is not merely an extension of the two-body problem but possesses *significantly* different properties.

In what follows, we will merely outline Poincaré's approach on an elementary basis. For a deeper insight, the papers by Urs Kirchgraber in [3] and [4] are highly recommended.

Poincaré fundamentally considered the general three-body problem. However, he gained essential insights using the example of the restricted three-body problem. We want to examine this case, in which the two heavy masses are again to rotate in circular orbits around the centre of mass. First, we want to clarify the degrees of freedom of the system.

We choose the same notation and the same coordinate system as in the previous section. Then the two heavy masses are at rest in this system. We therefore need only describe the motion of the small mass μ. At a given time$t$, this mass is located at a point $\vec{r}(t) = \binom{x(t)}{y(t)}$. In doing so, the differential equations (7) must be satisfied.

Using a transformation such as the one in the school project "*Numerical Methods and Coupled Pendulums,*"

$$u(t) = \dot{x}(t), v(t) = \dot{y}(t)$$

, this system of equations is transformed into a system of four first-order differential equations:

$$(9) \begin{cases} \dot{x} = u \\ \dot{u} = \dfrac{-(1-\sigma)(1+x)}{\sqrt{(1+x)^2+y^2}^3} + \dfrac{(1+\sigma)(1-x)}{\sqrt{(1-x)^2+y^2}^3} + \dfrac{x-\sigma}{4} \\ \dot{y} = v \\ \dot{v} = \dfrac{-(1-\sigma)y}{\sqrt{(1+x)^2+y^2}^3} + \dfrac{-(1+\sigma)y}{\sqrt{(1-x)^2+y^2}^3} + \dfrac{y}{4} \end{cases}$$

We are looking for functions $x(t), u(t), y(t), v(t)$ such that this system of differential equations is satisfied. The motion takes place in a *four-dimensional* phase space (which we encountered, among other places, in the school project *"Numerical Methods and Coupled Pendulums"*). A trajectory of the mass μ in phase space is then a function

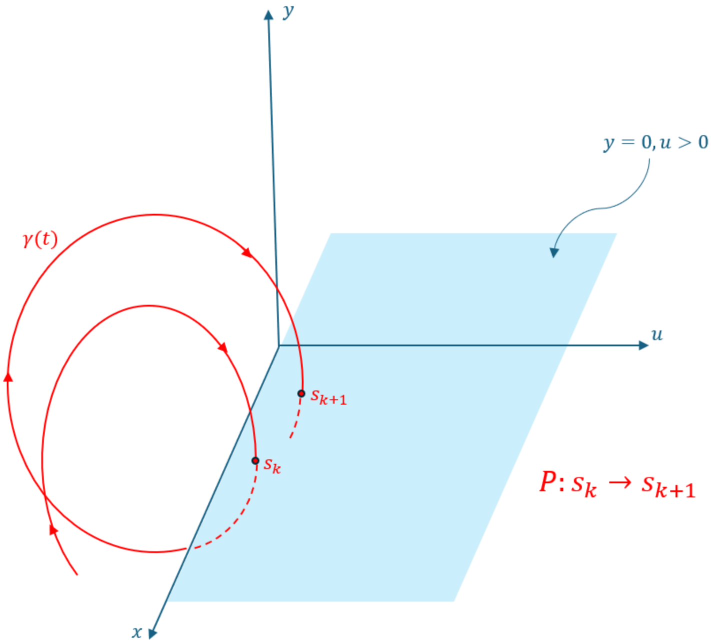
$$\gamma: t \in \mathbb{R} \to (x(t), u(t), y(t), v(t)) \in \mathbb{R}^4$$

Now the conservation laws apply. When choosing the coordinate system, we have already taken advantage of the fact that the total momentum is constant and that the centre of mass can be assumed to be at rest. The conservation of angular momentum implies that the trajectories lie in a plane. This leaves the conservation of total energy. This means: If $x(t), u(t), y(t)$ are known, then $v(t)$ is determined. Thus, the system has *three* degrees of freedom. It is therefore sufficient to consider trajectories in *three-dimensional* space.

An equilibrium position of the system is a "trajectory" consisting of only a fixed point. We determined such positions in the previous section. The next step would now be to determine *periodic* trajectories, i.e., trajectories for which the following holds:

$$\exists \tau: \gamma(t+\tau) = \gamma(t), \forall t$$

This seems extraordinarily difficult. But here Poincaré makes a first decisive approach: He reduces a periodic trajectory to the fixed-point problem by introducing a plane transverse to the trajectory and then considering the intersections of the trajectory with this plane. This procedure is also known as the "Poincaré section" and is used in the simulator for some simulations. The following sketch illustrates the idea:



Sketch of the Poincaré section

Poincaré now considers only the points of intersections_k of the trajectory with a transverse plane, in the sketch the plane $y = 0, u > 0$. This defines a mapping:

$$P: s_k \to s_{k+1} =, \mathbb{R}^2 \to \mathbb{R}^2$$

This achieves the following:

1) Instead of the continuous system (9) and the sought-after trajectories $\gamma(t)$, we have—with a suitable choice of a transverse section plane—a *discrete* system defined by the mapping P
2) The dimension is reduced by one: P operates on $\mathbb{R}^2$
3) Periodic trajectories in the three-dimensional case are periodic points of P or *fixed points of* P^m for a suitable $m \in \mathbb{N}$

We now examine a mapping P with the following notation:

$$P: \begin{pmatrix} x \\ y \end{pmatrix} \to \begin{pmatrix} P_1(x, y) \\ P_2(x, y) \end{pmatrix}$$

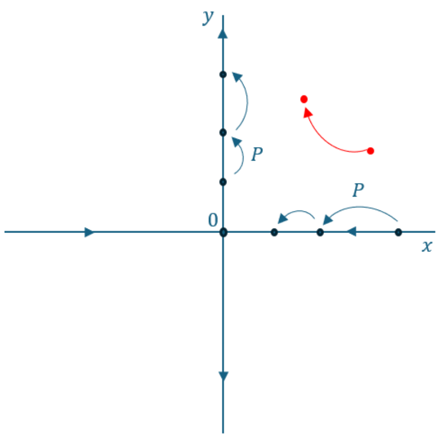
Let P also be a *diffeomorphism*, i.e., the inverse map P^{-1} exists, and both P and P^{-1} are continuously differentiable.

In the school project "*Strange Attractors and Edward Lorenz's Weather Forecast*," we explored what the fixed points of such a mapping might look like. Suppose that P has a fixed point and we choose this as the origin of the coordinate system. Then $P(\vec{0}) = \vec{0}$ holds. In this school project, we learned that the behaviour of the mapping near a fixed point is determined by the so-called Jacobi matrix, specifically by its eigenvalues. Here we write for this matrix:

$$dP := \begin{bmatrix} \frac{\partial P_1}{\partial x} & \frac{\partial P_1}{\partial y} \\ \frac{\partial P_2}{\partial x} & \frac{\partial P_2}{\partial y} \end{bmatrix}$$

In the aforementioned school project, we observed: If the eigenvalues of this matrix are complex, this leads to spiral-shaped orbits near the fixed point. For real eigenvalues, the fixed point can be attractive or repulsive in the direction of a corresponding eigenvector. If the following holds for both eigenvalues: $\lambda_{1,2} \in \mathbb{R}, 0 < |\lambda_{1,2}| < 1$, then the fixed point is attractive and the corresponding periodic orbit *is stable*.

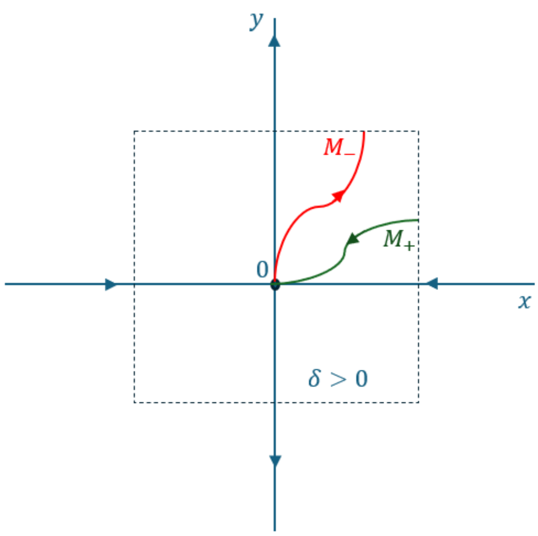
Now, the case of a *homoclinic* fixed point is interesting, where we have two real eigenvalues λ_1, λ_2 with $0 < \lambda_1 < 1 < \lambda_2$. This means that the fixed point is attractive in the direction of the first eigenvector and repulsive in the direction of the second. See the school project *"Strange Attractors" and Edward Lorenz's weather forecast*. We have the following situation when taking the eigenvectors as a basis (we draw them perpendicular to each other unless otherwise specified):



Behaviour near a homoclinic fixed point

Points that start precisely on the x-axis move toward the origin. Points that start on the y-axis move away from the origin. Points that do not start on the axes generally move like the point shown in red.

Poincaré showed that, in a small neighbourhood of the origin, two trajectories M_+, M_- can be constructed that intersect at the origin and lie *tangent* to the eigenvectors.



The local stable and unstable manifolds

The trajectories have the following properties:

Points starting on the trajectory M_+ move toward the origin and come arbitrarily close to it: $s \in M_+ \Longrightarrow \lim_{n\to\infty} P^n(s) = 0$. M_+ is also invariant under P , i.e., $P(M_+) \subseteq M_+$ holds. M_+ is called *a local stable manifold*.

Points starting on the path M_- move away from the origin. M_- is invariant under P^{-1}, i.e., $P^{-1}(M_-) \subseteq M_-$ holds. If one runs the iteration backward, i.e., iterates P^{-1}, points on M_- approach the origin arbitrarily closely: $s \in M_- \Longrightarrow \lim_{n\to\infty} P^{-n}(s) = 0$. M_- is also invariant under P^{-1}, i.e., $P^{-1}(M_-) \subseteq M_-$ holds. M_- is called *a locally unstable manifold*.

10. Transverse homoclinic points

Now we follow the paper by Urs Kirchgraber in [3] and outline the further procedure. The trajectory M_- is "traced forward" and the trajectory M_+ "traced backward." We form:

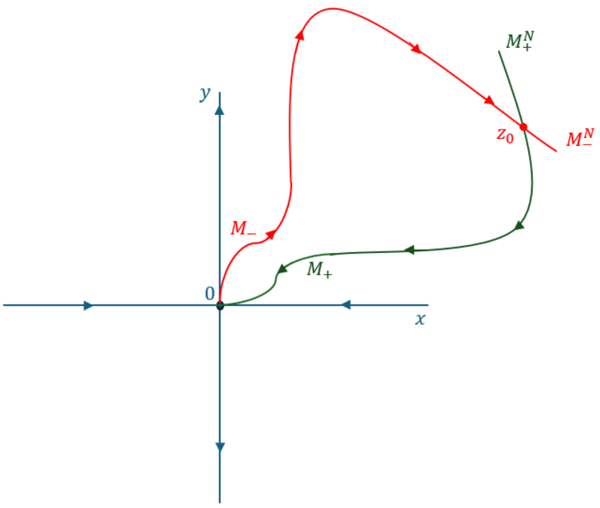
$$M_-^N = P^N(M_-), M_+^N = P^{-N}(M_+)$$

In an exercise, it is shown that the following holds:

1) $M_-^N \subseteq M_-^{N+1}, M_+^N \subseteq M_+^{N+1}$
2) M_-^N is invariant under P^{-1}. M_+^N is invariant under P.
3) $s \in M_+^N \Longrightarrow \lim_{n\to\infty} P^n(s) = 0$. $s \in M_-^N \Longrightarrow \lim_{n\to\infty} P^{-n}(s) = 0$

This means that the properties of M_- and M_+ carry over to M_-^N and M_+^N , respectively.

It is possible that M_-^N and M_+^N intersect again outside the origin, and indeed transversally. We then have the following situation:



M_-^N and M_+^N intersect transversally at the point z_0

The point z_0 is then called *a transversal homoclinic intersection.* This term goes back to Poincaré. He recognized that the behaviour of the system is extraordinarily complex when such an intersection exists. The two manifolds M_-^N and M_+^N then intersect infinitely many times as N continues to grow, and in a very complex manner.

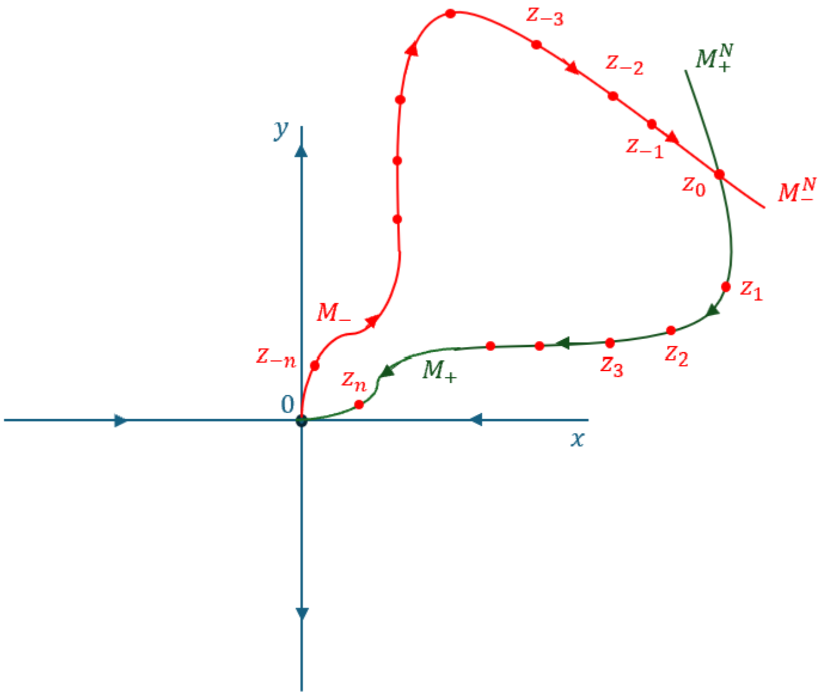
Our goal is to outline later that this system has chaotic properties, summarizing the results of Urs Kirchgraber in [3].

We start from the intersection pointz_0 and run the iteration P forward and backward. Then we obtain a set of points:

$$\{z_i |\, z_i \coloneqq P^i(z_0), i \in \mathbb{Z}\}$$

From the construction of M_-^N and M_+^N, it follows that:

1) $z_i \in M_-^N$ for $i \leq 0$ and $z_i \to 0, i \to -\infty$
2) $z_i \in M_+^N$ for $i \geq 0$ and $z_i \to 0, i \to \infty$



The points $z_i \coloneqq P^i(z_0),\ i \in \mathbb{Z}$ on M_-^N or M_+^N

0 is a cluster point (the only one) of the set of points $\{z_i | i \in \mathbb{Z}\}$, and we consider the set

$$\Lambda \coloneqq \{0\} \cup \{z_i |\, i \in \mathbb{Z}\}$$

Since M_-^N and M_+^N intersect at z_0, there is a *pair* of tangents at this point to M_-^N and M_+^N; we call them t_0^- and t_0^+. As shown in [3], a pair of tangents $t_i^{\pm}$ can also be “attached” to any other point z_i. At the origin, the eigenvectors of $dP(0)$ were tangent to M_- and M_+ , respectively, due to the construction of $M_{\pm}$. We denote these eigenvectors as t_∞^- and t_∞^+ , respectively. In [3], it is shown that this field of two-vectors can be constructed such that:

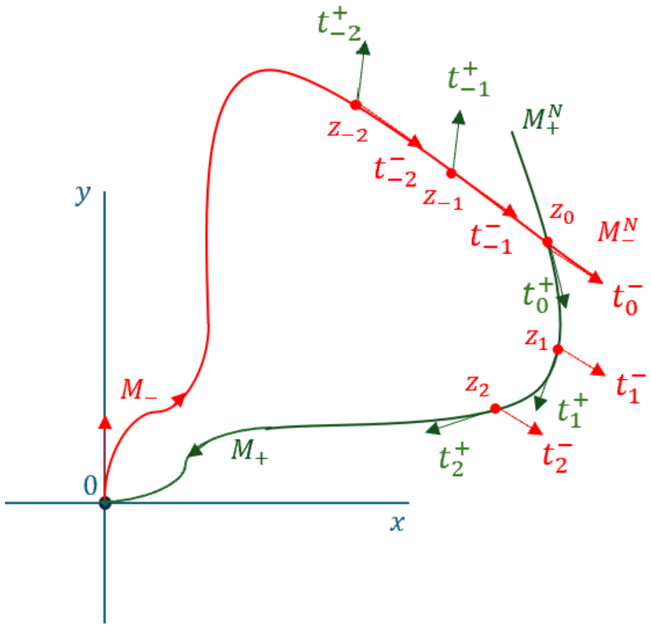
3) t_i^- is tangent to M_-^N and transverse to M_+^N. t_i^+ is tangent to M_+^N and transverse to M_-^N
4) The field of bipeds *is invariant under the tangent map* $dP(z_i)$, that is: $dP(z_i)t_i^{\pm}$ is parallel to $t_{i+1}^{\pm}$ for $i \in \mathbb{Z}$.
5) dP is *contracting* in the direction of t_i^+ and *expanding* in the direction of t_i^- .
6) The field of bipeds *is continuous,* and the following holds: $\lim_{i\to-\infty} t_i^{\pm} = t_\infty^{\pm}$ and $\lim_{i\to\infty} t_i^{\pm} = t_\infty^{\pm}$

Regarding point 4): If we denote the proportionality factor by $\lambda_i^{\pm}$, then: $dP(z_i)t_i^{\pm} = \lambda_i^{\pm} t_{i+1}^{\pm}, i \in \mathbb{Z}$.
Regarding point 5): There exists a number $\vartheta \in (0,1)$ such that: $|\lambda_i^+| < \vartheta$ and $|\lambda_i^-| > \frac{1}{\vartheta}$ for $i \in \mathbb{Z}$.

Definition 10.1

The set Λ constructed in this way, together with properties 1) through 6), is a *hyperbolic set*. □

This set plays an important role in the next section.



The field of bipeds

11. The Poincaré–Smale Theorem

The goal of this section is to show that the dynamical system consisting of the hyperbolic set Λ together with the iterated P^m is chaotic for a suitable $m \in \mathbb{N}$. In the school project *"The Chaotic Properties of Logistic Growth*," we noted that the logistic growth is conjugate to the tent map for the parameter $a = 4$. Urs Kirchgraber uses a similar approach in [3], and we will outline it here.

First, some preliminaries. Instead of the tent map, we use here the set Σ of all two-sided infinite 0-1 sequences together with the Bernoulli shift map σ, which shifts such a sequence one place to the left:

$$\Sigma = \{(\dots, s_{-2}, s_{-1}; s_0, s_1, \dots)|s_i \in \{0,1\}\}$$

$$\sigma(\dots, s_{-2}, s_{-1}; s_0, s_1, \dots) = (\dots, s_{-2}, s_{-1}, s_0; s_1, \dots)$$

We further define a metric on Σ:

$$d(s, s') = \max\{2^{-|i|}|s_i \neq s'_i, i \in \mathbb{Z}\}$$

This makes Σ a (compact) metric space. The dynamical system (Σ, σ) is obviously chaotic in the sense of the coin toss.

As further preparation, we need the *shadow trajectory lemma*. This goes back to the mathematicians Dmitri Anosov (1936–2014) and Rufus Bowen (1947–1978).

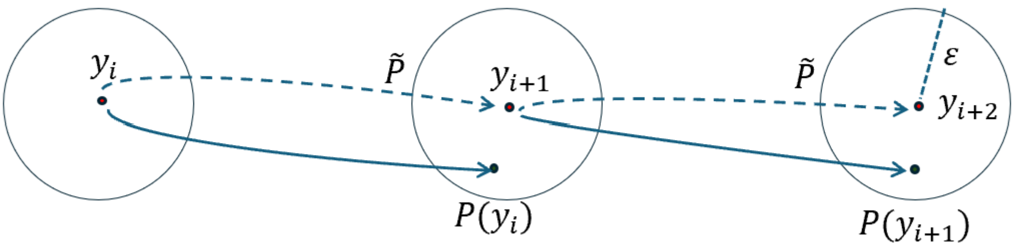
A pseudo-orbit is a “near-orbit” of an iteration, that is, an orbit that is slightly flawed. The shadowing lemma then states that, under certain conditions, for every pseudo-orbit there exists a “true” orbit that runs arbitrarily close to the pseudo-orbit. This is the shadow orbit of the pseudo-orbit. In this respect, pseudo-orbits are traces of real orbits. This is also interesting regarding the simulations in the simulator, because the simulated orbits on are indeed subject to a certain error due to the approximation methods used. In the chaotic case, however, these simulated and slightly erroneous orbits can be traces of real orbits. We will elaborate on this further below.

Definition 11.1

We consider the dynamical system (Λ, P) as in the previous section.

A sequence $\{y_i | y_i \in \Lambda, i \in \mathbb{Z}\}$ is called *an ε-pseudo-orbit* if the following holds: $|y_{i+1} - P(y_i)| \le \varepsilon, \forall i \in \mathbb{Z}$. □

For a real orbit, $y_{i+1} = P(y_i)$ would hold. A pseudo-orbit is thus slightly off a real orbit at every step.



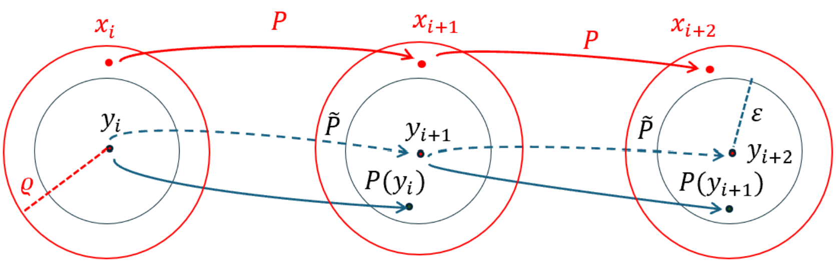
Instead of the correct iteration P, one uses the (slightly) incorrect $\tilde{P}$

For many dynamic systems described by a system of differential equations, numerical approximation methods are used to obtain a picture of their dynamics. In other words, instead of the correct iteration P, a slightly incorrect one $\tilde{P}$ is used. At each step, however, the error remains smaller than ε. Nevertheless, the error accumulates, and the question is whether the orbit at $\tilde{P}$ still represents a "true" orbit as described at P. The answer is given by the following shadowing lemma: For every pseudo-orbit, there is *exactly one* "real" orbit that remains close to the pseudo-orbit.

Lemma 11.2 (*Shadowing Lemma*)

Let Λ be a hyperbolic set and$P: \Lambda \to \Lambda$ the corresponding diffeomorphism.

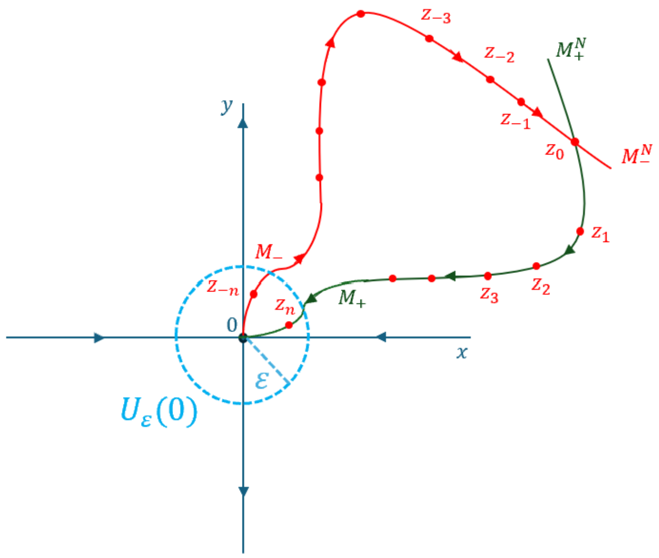
Claim: Then, for every sufficiently small $\varrho > 0$, there exists an $\varepsilon > 0$ such that every ε-pseudo-orbit $\{y_i\}$ with $|y_{i+1} - P(y_i)| < \varepsilon, i \in \mathbb{Z}$ is ϱ —shadowed by *exactly one real orbit*. That is, there exists exactly one orbit $\{x_i\}$ such that $|x_i - y_i| < \varrho, i \in \mathbb{Z}$. $\{x_i\}$ is called *the ϱ -shadow-orbit* of the ε-pseudo-orbit $\{y_i\}$. □



Sketch of the shadowing lemma

Proving this lemma is beyond our capabilities. However, in [3] one can find a proof sketch that, with a little effort, is accessible and understandable to those with a basic understanding.

Now we are ready to outline how the dynamic systems (Λ, P) and (Σ, σ) are related. We follow the construction by Urs Kirchgraber in [3].



Outline of the construction of the point sets Y_0 and Y_1

We consider a small ε-neighbourhood of 0. If we choose $n \in \mathbb{N}$ large enough, then both z_{-n} and z_n lie in this neighbourhood due to properties 1) and 2) in the previous section. Furthermore, $P^{2n+1}(z_{-n}) = z_n$.

Now let us consider two sequences of points, each consisting of $m \coloneqq 2n + 3$ points:

$$Y_0 := 0, 0, \ldots\ldots, 0, 0$$

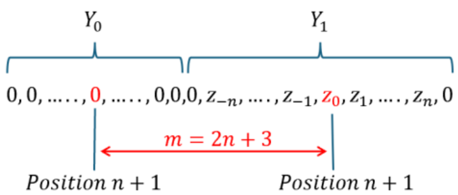
$$Y_1 := 0, z_{-n}, z_{-n+1}, \ldots, z_{-1}, z_0, z_1, \ldots, z_{n-1}, z_n, 0$$

The first sequence is *an orbit* under the iteration P, since $P(0) = 0$. The second sequence is *almost an orbit* under the iteration P. This is because we make an error in the first step $0 \longmapsto z_{-n}$ and in the last step $z_n \longmapsto 0$. This error is smaller than ε. The second sequence is therefore an ε-pseudo-orbit. We also note that at the position $n + 1$, the value is 0 forY_0 and z_0 for Y_1. If we combine the two sequences, for example Y_0Y_1 , then these positions are separated by $m = 2n + 3$.

Now let

$$\{y_i\} = \cdots Y_0Y_1Y_1Y_0Y_1Y_0 \ldots$$

a sequence consisting of an arbitrary combination of the sequences Y_0 and Y_1. We consider the positions $i = N + 1 + km, k \in \mathbb{Z}$ in this sequence. As previously defined, $m = 2n + 3$.



Combination of Y_0 and Y_1

We now take an arbitrary two-sided infinite 0-1 sequence

$$s = (\dots, s_{-3}, s_{-2}, s_{-1}; s_0, s_1, s_2, s_3, \dots.) \in \Sigma$$

and then replace every 0 in this sequence with Y_0 and every 1 with Y_1. This gives us a sequence of points that represents a pseudo-orbit:

$$(y_i) = (\dots, Y_{s_{-3}}, Y_{s_{-2}}, Y_{s_{-1}}; Y_{s_0}, Y_{s_1}, Y_{s_2}, Y_{s_3}, \dots \dots)$$

Then, for each Y_i, we consider only the position $n + 1$. These positions for two adjacent Y_i have the distance m. These are then all the positions at the points $n + 1 + im, i \in \mathbb{Z}$ of the sequence (y_i). This gives us a "stroboscopic" image of the pseudo-orbit (y_i) , which consists of either the point 0 or the point z_0. Such a sequence might look like this:

$$(\tilde{y}_i) = (\dots, 0, 0, z_0, 0; z_0, z_0, 0, \dots..)$$

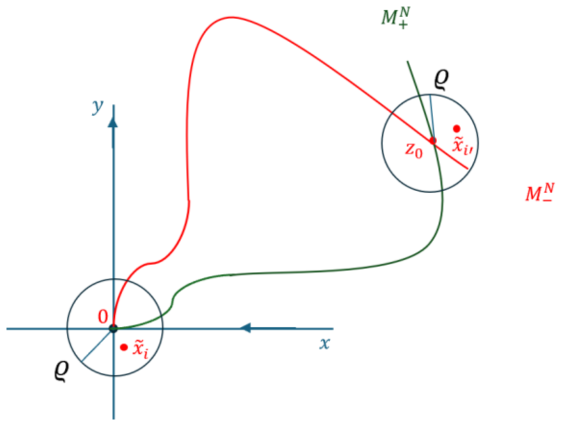
And the following holds:

$$\tilde{y}_i = \begin{cases} 0 \Leftrightarrow s_i = 0 \\ z_0 \Leftrightarrow s_i = 1 \end{cases}$$

Now, according to the shadowing lemma, for every pseudo-orbit (y_i) there is exactly one (true) shadowing orbit (x_i) , such that $|y_i - x_i| < \varrho$ for a small $\varrho > 0$. If we again consider only the positions $n + 1 + im, i \in \mathbb{Z}$ from this shadowing orbit, and denote this stroboscopic sequence by $(\tilde{x}_i)$, then the following holds:

$$\tilde{x}_i \in \begin{cases} K(0, \varrho) \Leftrightarrow s_i = 0 \\ K(z_0, \varrho) \Leftrightarrow s_i = 1 \end{cases}$$

If $K(0, \varrho)$ and $K(z_0, \varrho)$ denote the circles around 0 and z_0, respectively, with radius ϱ.



Positions of the elements of the shadow path $(\tilde{x}_i)$

For every 0-1 sequence $s \in \Sigma$, we have thus been able to assign a unique (stroboscopic) shadow orbit $(\tilde{x}_i)$ with the properties mentioned above through this construction. If we denote the set of all shadow paths obtained in this way by $\tilde{X} \subset \mathbb{R}^2$, we have a bijective mapping:

$$\tau: s \in \Sigma \to \tau(s) = (\tilde{x}_i) \in \tilde{X}$$

The iterated mapping P^m shifts the sequence $(\tilde{x}_i)$ one place to the left, just as the Bernoulli shift mapping σ acts on a sequence in Σ. Thus, the following holds:

$$\tau \circ \sigma = P^m \circ \sigma$$

And it follows that:

Theorem 11.3

Assumption: Let $\Lambda \subset \mathbb{R}^2$ be a hyperbolic set and $P: \Lambda \to \Lambda$ the corresponding diffeomorphism.

Claim:

1) The systems $(\tilde{X}, P^m)$ and (Σ, σ) constructed earlier are *conjugate*
2) The system $(\tilde{X}, P^m)$ is chaotic in the sense of the coin toss.

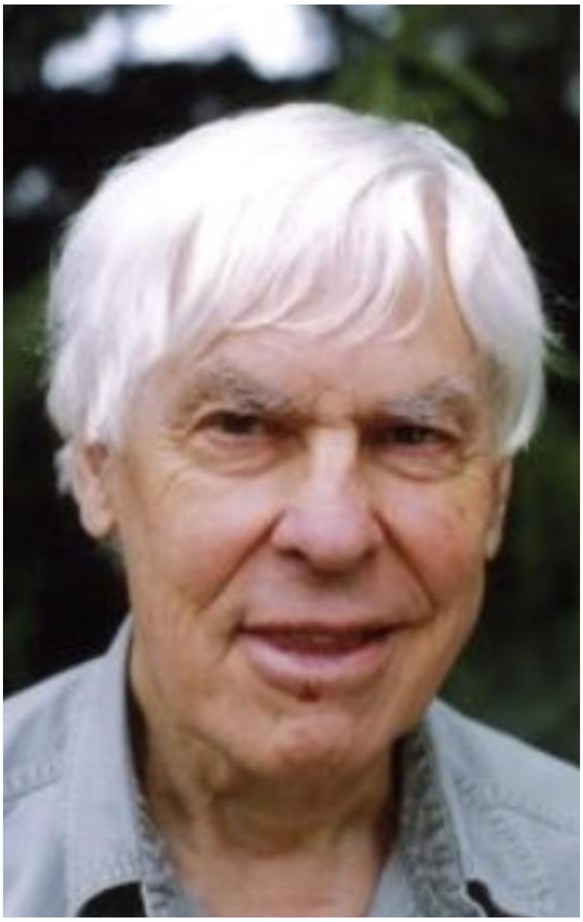
□

Proof: See the previous construction. By the assumption, the shadow path lemma holds. It follows that 1). By conjugation, the chaotic properties of the system (Σ, σ) are transferred to $(\tilde{X}, P^m)$.

□

In Urs Kirchgraber's paper [3], one finds a detailed derivation of this result as well as a proof sketch of the shadowing lemma. Urs Kirchgraber also provides a concrete example with an application in the form of a two-dimensional system of ordinary differential equations.

Theorem 11.3 is a special case of the Poincaré–Smale theorem, which is mentioned here as a future prospect.



Stephen Smale (1930–…) is an American mathematician. For his 1961 proof of the Poincaré conjecture in the case $n \geq 5$, he received the Fields Medal, the highest honour in mathematics and comparable to the Nobel Prize in other sciences.

In the case of a dynamical system with a hyperbolic equilibrium and a transverse homoclinic point, he recognized—as Henri Poincaré had before him—that the associated stable and unstable manifolds intertwine in a complex manner an infinite number of times.

With his famous horseshoe construction, he proved the Poincaré–Smale theorem.

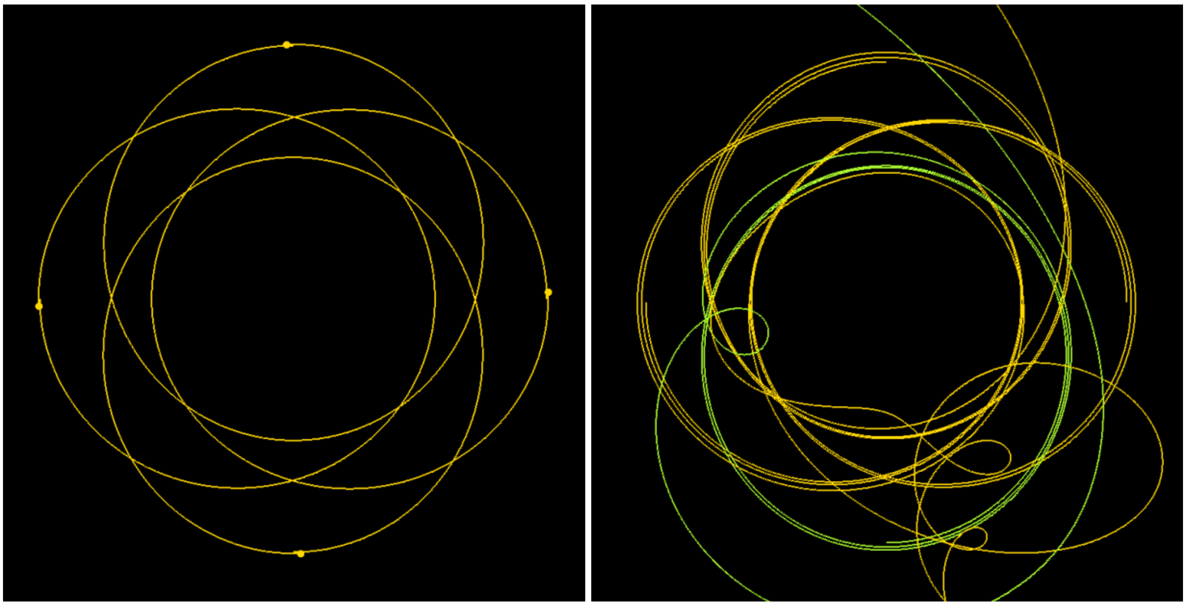
Theorem 11.4 (*Poincaré–Smale Theorem*)

Assumption: $P: \mathbb{R}^2 \to \mathbb{R}^2$ has a hyperbolic equilibrium and a homocline point transverse to it.

Claim: There exists a natural number m and a subset $S \subset \mathbb{R}^2$ invariant under P^m such that the discrete dynamical system (S, P^m) and the Bernoulli system (Σ, σ) are conjugate to each other. □

To wrap up, let's run a few simulations using the simulator. In certain configurations, we'll actually find that the system reacts sensitively to the initial conditions. Let's look at an example.

In the following simulation, four stars of equal mass orbit around a common centre of mass. The velocity at perihelion is identical in magnitude and tangent to the expected orbit, which is indicated at the start. If the system is symmetric (all stars are yellow), then the dynamics are periodic. However, if the distance to the centre of mass is altered even slightly, the system tips into chaos. If we take the distance of the yellow stars from the centre of mass as a unit, the distance for the green star in this case is 0.99999995. Let's run the experiment:



Left: Symmetrical situation. Right: Disturbed situation.

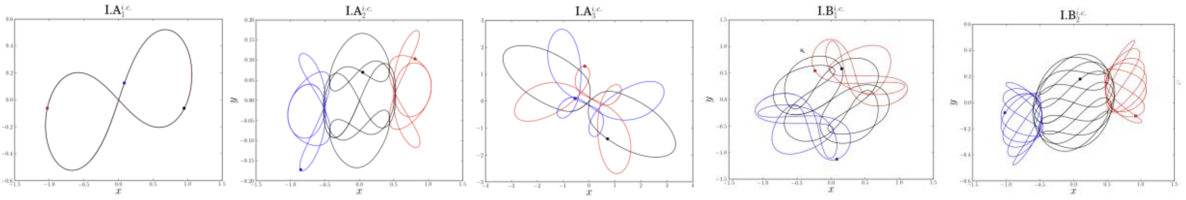
12. Periodic Orbits

An exciting topic in the (normalized) universe is the question of periodic orbits, particularly in the three-body problem. The first families of simple periodic orbits were discovered by Leonhard Euler (1707–1783) and later by Joseph Louis Lagrange (1736–1813). In the 1990s, computer-aided methods revealed additional, sometimes surprising, periodic orbits.

Examples can be found online at the following links:

https://numericaltank.sjtu.edu.cn/three-body/three-body.htm

https://numericaltank.sjtu.edu.cn/three-body/three-body-movies.htm



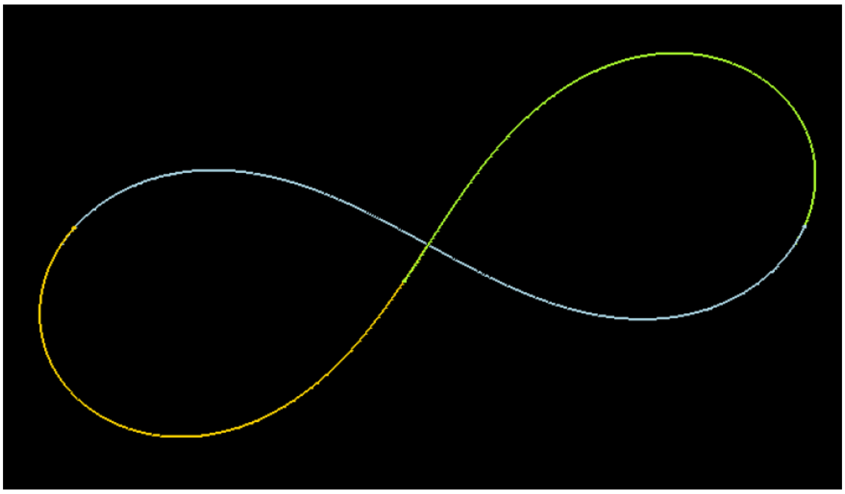
Examples of periodic stable orbits on the Internet

Further explanations can also be found in the links above. Some of these scenarios are saved as configurations in the simulator. In particular, the case of three stars with equal masses $m_1 = m_2 = m_3 = 1$ and the initial positions $\vec{r}_1 = (-1{,}0), \vec{r}_2 = (1{,}0), \vec{r}_3 = (0{,}0)$.

The initial velocities are: $\vec{v}_1 = (c_1, c_2) = \vec{v}_2, \vec{v}_3 = (-2c_1, -2c_2)$ where the values from $c_{1,2}$ can be found in the following table. Note that the total momentum at the start is zero.

The simulator includes several examples with low periods, in which the Runge-Kutta method clearly reaches its limits:

Constellation	C_1	C_2
IA1	0.3471168881	0.5327249454
IA2	0.3068934205	0.1255065670
IA3	0.6150407229	0.5226158545
IB1	0.4644451728	0.3960600146
IB2	0.4059155671	0.2301631260
IIC1	0.2827020949	0.3272089716



The Periodically Stable Orbit IA1

The Runge-Kutta method is not precise enough to represent the IA2 orbit: The orbit is unstable.

13. Appendix: Information about our solar system

Gravitational constant: $G \approx 6.6743 \cdot 10^{-11} \frac{m^3}{kg \cdot s^2}$

	Sun	Mercury	Venus	Earth	Mars
Mass in kg	1.9884E+30	3.3010E+23	4.8673E+24	5.9722E+24	6.4200E+23
Mass relative to the Earth's mass	332,943	0.055	0.815	1.000	0.107
Mass relative to the solar mass	1	1.660E-07	2.448E-06	3.004E-06	3.229E-07
Diameter in km	1,392,000	4,879	12,103	12,735	6,772
Gravitational constant in m/s²	274.00	3.70	8.87	9.80	3.73
Orbital period around the Sun in days	-	87.969	224,701	365,256	686,980
Major semi-axis in million km	-	57,909	108,200	149,600	227,990
Large half-axle in AE	-	0.3871	0.7233	1,000	1.524
Eccentricity of the elliptical orbit	-	0.2056	0.0068	0.0167	0.0934
Relative speed to the Sun in km/h	-	172,332	126,072	107,208	86,868
Relative solar speed in AU/day	-	0.0276	0.0202	0.0172	0.0139
Escape velocity in km/s *)	617.4	4.3	10.4	11.2	5.0
Argument of perihelion **)	-	1.3519	2.2956	1.7967	5.8650
Perihelion in AU	-	0.3075	0.7184	0.9833	1.3814
Perihelion speed in km/h	-	212,328	126,936	109,044	95,400
Perihelion speed in AU/day	-	0.034063	0.020364	0.017494	0.015305
Aphelion in AU	-	0.467	0.728	1.017	1.666

	Jupiter	Saturn	Uranus	Neptune
Mass in kg	1.8980E+27	5.6830E+26	8.6800E+25	1.0024E+26
Mass relative to Earth's mass	317.8	95.2	14.5	16.8
Mass relative to the solar mass	9.545E-04	2.858E-04	4.365E-05	5.041E-05
Diameter in km	138,346	114,632	50,532	49,105
Gravitational constant in m/s²	24.79	10.44	8.87	11.15
Orbital period around the Sun in days	4,329	10,751	30,664	60,148
Semi-major axis in million km	778.51	1,433.40	2,872.40	4,514.60
Major semi-axis in AU	5.204	9,582	19,201	30,178
Eccentricity of the elliptical orbit	0.0489	0.0542	0.0472	0.0097
Relative speed to the Sun in km/h	47,052	34,884	24,516	19,548
Relative solar speed in AU/day	0.0075	0.0056	0.0039	0.0031
Escape velocity in km/s *)	60.2	36.1	21.4	23.6
Argument of perihelion **)	0.2575	1.6132	2.9839	0.7849
Perihelion in AU	4.9501	9.0481	18.3755	29.7667
Perihelion speed in km/h	49,392	36,648	25,596	19,800
Perihelion speed in AU/day	0.007924	0.005879	0.004106	0.003176
Aphel	5.455	10.12	20.11	30.069

*) at the equator

**) Relative to the zero direction, which is defined by the intersection of the ecliptic with the Earth's equatorial plane. The value is given in radians.

The data on the perihelion change over time. The above data are from the year 2000.

14. Exercise examples

1. We are trying to approximate an ellipse with semi-axes a, b using an epicycle. This is a curve of the form:
$$\vec{r}(t) = R\begin{pmatrix}\text{cos}t\\ \text{sin}t\end{pmatrix} + r\begin{pmatrix}\cos(\omega t)\\ \sin(\omega t)\end{pmatrix}$$
where R is the radius of the large circle and r is the radius of the small circle, whose centre moves along the large circle.
We want this curve to touch the ellipse at its foci. How must the parameters R, r, ω be chosen? For which a, b is the approximation valid? For which b does the epicyclic curve begin to form loops?

2. Two-body problem: Calculate the trajectory of the mass m_1 relative to the position of the mass m_2, rather than relative to the centre of mass. Compare the results, in particular for the values of p, ε.

3. Determine the equation of the conic section in Cartesian coordinates from its polar form in Theorem 5.2.

4. The gravitational constant is known. The appendix “Information on Our Solar System” also provides the most important data from this system.

   a) Using this information, calculate the orbital period T of the Earth around the Sun.

   b) Calculate the angular momentum of the Earth–Sun system.

   c) Calculate the explicit polar equation of the Earth’s orbit around the Sun.

   d) The orbital period of Mars around the Sun is 687 Earth days. Determine the semi-major axis of Mars’s orbit from this information.

5. Investigate the N-body problem analogously to the three-body problem. What do the differential equations look like? Show that the system is independent of the choice of origin and the direction of the coordinate system. Do the conservation laws hold?

6. Investigate symmetric initial positions in the “Simulator.” That is, the bodies under consideration all have the same mass, and their initial positions and initial velocities are rotationally symmetric about the origin.

7. Determine the equation of Jupiter’s orbit around the Sun in astronomical units.

8. A body lies on an ellipse with major axis $a = 2$ and minor axis $b = 1$. At the start, the body is at the positions $\vec{r}_0 = \binom{0.5176}{1.9318}$ and $\vartheta_0 = 1.309$. Determine the equation of the ellipse’s orbit.

9. Determine the orbital equation of Jupiter in AU. Use this to calculate Jupiter’s aphelion and perihelion. Compare the result with data from the Internet. Note that the result is only an approximation.

10. If the Earth were to move in a circular orbit around the Sun, the gravitational force and the centrifugal force would exactly cancel each other out. What would the Earth’s speed be in km/h? Compare the result with the speed listed in the table about the solar system.

11. Using the data in the appendix, determine the centre of mass of our planetary system if all the planets were aligned on the same straight line through the Sun.

12. Show that the following holds for the three-body problem: $\frac{d}{dt}\left(E_{kin} + E_{pot}\right) = 0$

13. Investigate the existence of solutions for the Lagrange points L_2 and L_3 in Theorem 8.3.

14. Theorem 8.3: In the case $\sigma = 0$, compute the equilibrium position L_1 explicitly. Furthermore, show that L_2 and L_3 are symmetric about the origin.

15. Consider the local stable and unstable manifolds M_+, M_- as in Section 9. Prove:

   1) $M_-^N \subseteq M_-^{N+1}, M_+^N \subseteq M_+^{N+1}$

   2) M_-^N is invariant under P^{-1} . M_+^N is invariant under P.

1) $s \in M_{+}^{N} \Longrightarrow \lim_{n\to\infty} P^{n}(s) = 0.\ s \in M_{-}^{N} \Longrightarrow \lim_{n\to\infty} P^{-n}(s) = 0$

16. Show that the dynamical system (Σ, σ) from Section 11 is chaotic in the sense of the coin toss and also in the sense of Devaney.

Further Reading

[1] Orbital Motion, Archie E. Roy, Taylor & Francis Group, 2005

[2] Celestial Mechanics, John M.A. Danby, Willmann-Bell Inc., 1992

[3] Urs Kirchgraber: Chaotic Behaviour in Simple Systems, Reports on Mathematics and Teaching, ETHZ, 1992

[4] Urs Kirchgraber: When Poincaré, Hadamard, and Perron Discovered Invariant Manifolds, Reports on Mathematics and Education, ETHZ, 1995